國立陽明交通大學

資訊科學與工程研究所

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National Yang Ming Chiao Tung University

Master's Thesis

Compile Once, Execute Many：重複性模板實例化任務的可驗證規格編譯方法

Compile Once, Execute Many: Verifiable Specification Compilation for Recurrent Template-Instantiation Tasks

研 究 生： 林冠丞 (Lin, Kuan-Cheng)

指導教授： 陳添福 (Chen, Tien-Fu)

中華民國 一一五年十一月

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摘　要

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Student : Kuan-Cheng Lin Advisor: Dr. Tien-Fu Chen

Institute of Computer Science and Engineering
National Yang Ming Chiao Tung University

Abstract

(Abstract to be written.)

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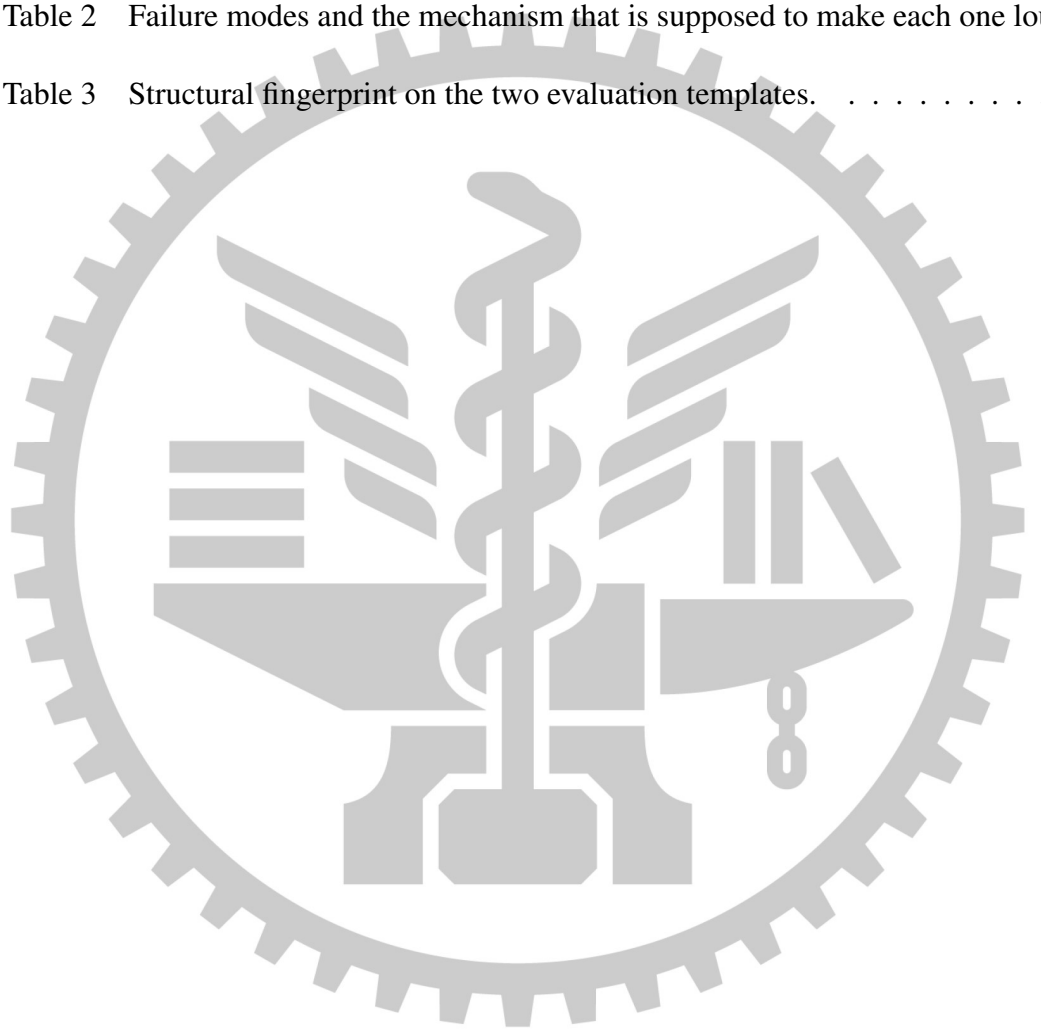

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3.1 Problem Formulation

3.1.1 A Stateful, Two-Stage Input Model

The abbreviation $\mathcal{A} : T \to S$ is potentially misleading because the system does not consume all experimental material in one call, and derivation is neither pure nor deterministic. Let d denote one derivation dataset in template lineage ℓ. We split the available material into a derivation layer and a later regression layer,

$$T = \big(T^{(n)}_{\mathrm{der},d}, T_{\mathrm{reg},\ell}\big), \tag{3.1}$$

$$T^{(n)}_{\mathrm{der},d} = \big(d, \ell, B_d, Y^?_d, \{X_{dj}\}_{j=1}^{N_d}, R^?_d, U^?_d, Q_d, P, \Gamma, \Theta, H_d^{(<n)}\big), \tag{3.2}$$

$$T_{\mathrm{reg},\ell} = \mathcal{R}_\ell = \big\{c_i = \big(I_i, \{X_{ij}\}_{j=1}^{N_i}, u_i, \pi_i, Y_i^?\big)\big\}_{i=1}^{K_\ell}. \tag{3.3}$$

Here, B_d is exactly one baseline template; $Y^?_d$ is an optional expected output; $\{X_{dj}\}_{j=1}^{N_d}$ is a possibly empty collection of source artifacts; $R^?_d$ is an optional operational-rule artifact; and $U^?_d$ is an optional inline source-data override. When present, the override bypasses the locally parsed artifact dictionary and, on the agent path, takes precedence over extracted keys. Q_d groups the inspection level, force-refresh choice, content-addressed dump, cache state, CAD preflight and health observations, locator verification, and deterministic validation/repair feedback. The derivation environment is also an input: P denotes the prompt files, Γ the agent's

tool and skill definitions, and Θ the model, gateway, and MCP configuration. Thus, changing a prompt, an allowed tool, or the selected model changes T_{der} even when all customer artifacts are byte-identical.

The term $H_d^{(<n)}$ is the pre-existing state before the nth derive. It contains the field names and rule inventories accumulated from all existing non-degraded specs derived from the same dataset. The implementation reads this state through `derivation_service.DerivationService._field_hints`, and `_previous_rule_inventory`; the degraded flag used by that selection is produced by `cad_health.degradation`. The hints are fed to the agent and the prior inventory is merged into the new spec evidence. With ω_n denoting model sampling and the observed session/tool trace, the honest derivation relation is therefore

$$S_d^{(n)} \sim \mathcal{A}\Big(T_{\mathrm{der},d}^{(n)}; \omega_n\Big), \qquad T_{\mathrm{reg},\ell} \notin \mathrm{dom}(\mathcal{A}). \tag{3.4}$$

Consequently, two derives over the same artifact bytes need not have the same context or the same output: $H_d^{(<n)}$ may have changed, and ω_n is stochastic. In this thesis, $\mathcal{A} : T \to S$ is only shorthand for the projection $\mathcal{A} : T_{\mathrm{der}} \rightsquigarrow S$ in Equation 3.4, not a claim that $\mathcal{A}$ is a pure function.

This split follows the implemented call boundary. The dataset creation and derive paths, `api.create_dataset` and `derivation_service.DerivationService.derive`, admit one baseline, at most one expected artifact, N_d source artifacts, and at most one rules artifact. They do not read `models.DwgTestCase`. Only after S exists does the separate endpoint call `regression.RegressionRunner.run`, which selects every active test case of the spec's lineage and executes them. K_ℓ is therefore the number of active cases at regression time; the implementation imposes no $K_\ell = 3$ limit. Each case c_i contains an input template I_i, its own source artifacts and explicit source-data override u_i, runtime parameters π_i, and an optional expected output $Y_i^?$.

The prompt, tools, and model terms above are concrete inputs rather than abstract nuisance variables. `derivation_agent.OpenCodeAgent.derive_rules` selects the `dwg-derivation` agent and `OpenCodeAgent._run` submits the task payload; the selected agent is configured by `opencode.jsonc`, its prompt is `prompts/derivation.md`, and its permitted CAD MCP tools are defined by the agent configuration and `mcp_server.server`. These files and settings must therefore be versioned with the material definition used in an experiment.

We reason about the resulting specification as

$$S = \big(G, L_s, L_t, F, \Pi, \mathcal{E}_S\big), \tag{3.5}$$

where G contains value-source and transformation semantics, L_s contains source locators, L_t contains independently derived target locators, F contains rendering rules, Π is the runtime-parameter schema, and $\mathcal{E}_S$ is derivation and validation evidence. The persisted `DwgSpec` stores these concerns across `field_map`, rule projections, `runtime_params`, and `evidence`; this tuple is a conceptual separation rather than a claim that each component is a distinct database column.

Information boundary. The operational-rule artifact R_d may identify source fields and state formulas, conditions, constants, and formatting requirements, but it must not contain target cell addresses, bookmark names, content-control tags, or mappings of the form `source_field` $\rightarrow$ `target_cell`. Target references belong to L_t and must be derived from the template by $\mathcal{A}$; otherwise R_d becomes an answer table and the derivation problem is leaked into its input. **Status: normative/aspirational only.** The boundary is stated but not enforced, in two independent ways. First, no admission check exists: neither `api.create_dataset`

nor `derivation_service.DerivationService._load_rules` rejects an answer-bearing rules artifact, so a hand-authored R_d containing target addresses is accepted silently. Second, the persisted projection is not structurally safe: `derivation_service.DerivationServic` copies the field's target-reference slot into `DwgSpec.mapping_rules`. That slot is carrier-agnostic — its interpretation follows the target kind of the template, so it holds bookmark identities for `.docx` and cell addresses for `.xlsx`, which are exactly the two artefact classes this boundary forbids. (The field retains the legacy name `handles` from the drawing-oriented prototype; the naming is historical and does not narrow the scope.) If that projection is exported, presented as the rule table, or fed back as R_d, it exposes target answers, and $\mathcal{A}$ is then measured on its ability to copy them rather than to derive them. It must therefore remain internal, and be stripped of target references or split from the semantic rules, before the boundary may be described as a property of the system rather than a requirement on it.

3.1.2 Compile-Once, Execute-Many

The target problem class consists of recurrent template-instantiation tasks for which the output is deterministically derivable from structured source data and explicit operational rules. "Compile once" means that the expensive, stochastic derivation in Equation 3.4 produces one reusable S; it does not mean that a single fixed output job is reused, because instance values differ. For instance input $V_i = (I_i, \{X_{ij}\}, u_i, \pi_i)$, execution deterministically instantiates and

applies the spec,

$$(J_i, P_i) = \mathcal{C}(S, V_i), \tag{3.6}$$

$$D_i = \mathcal{X}(J_i, I_i), \tag{3.7}$$

$$\mathrm{ok}_i = \mathcal{O}_{\mathrm{plan}}(P_i, I_i, D_i) \wedge \begin{cases} \mathcal{O}_{\mathrm{expected}}(D_i, Y_i), & Y_i^{?} = Y_i, \\ \mathrm{true}, & Y_i^{?} \text{ is absent.} \end{cases} \tag{3.8}$$

$\mathcal{C}$ is the deterministic per-instance compiler, P_i is its authoritative write plan, $\mathcal{X}$ is the format-specific executor, and D_i is the produced document. The problem is to derive an S once such that ok_i holds for every admissible instance, while the cost of derivation is amortized over repeated executions.

The deterministic core is implementation-backed. In `compiler.compile_cad_job`, the spec, source data, runtime parameters, and current template values are explicit inputs; failures are returned before submission, and the canonical job JSON is content-hashed. The DOCX/XLSX path shares this compiler and applies P_i through `docx_target.DocxTargetAdapter.` or `xlsx_target.XlsxTargetAdapter.apply`; `document_production.document_job` records a canonical hash for the document job. Regression is also implementation-backed: `regression.RegressionRunner._run_case` compiles each case, and `regression.Regress` uses `verification.verify_output` plus `verification.compare_expected` when Y_i exists.

This formulation deliberately localizes randomness to $\mathcal{A}$, and the execution path now matches that localization. An earlier revision of this chapter recorded that it did not: `production.ProductionC` invoked `derivation_agent.OpenCodeAgent.repair_cad_job` after a failed gate-way job validation, and the comparison that guarded the repaired job excluded target locations,

so a model-relocated target could be accepted and submitted inside the same run. That branch has since been removed from execution. A compiled job the tenant manifest rejects terminates the run with `CAD_JOB_MANIFEST_REJECTED` and an instruction to re-derive, review, and republish; a cached target reference the current template no longer contains terminates it with `CACHED_HANDLE_NOT_FOUND`. Neither is repaired in place, and `repair_cad_job` has no remaining caller on the execution path.

Status: implementation-backed, and pinned by a negative control rather than by inspection. `test_successful_production_operation_records_zero_llm_calls` replaces both agent repair entry points with a recorder that returns a failure, runs a complete production operation, and then asserts both that the recorder was never invoked and that the `LlmOperation` persisted for the run holds zero model calls. Reading the source would only show that the call site is absent today; the test also fails if a future change reintroduces one. This is the evidence behind claim C4, and it is deliberately the narrowest of the thesis's claims: it covers $\mathcal{X}$, not $\mathcal{A}$, and not the derivation-time repair loop of Chapter 4, which is by construction inside the compile step whose cost repeated execution amortizes.

3.1.3 Required Properties of the Specification IR

The reusable IR must satisfy the following five properties. Each status below refers to the repository state examined for this thesis, not merely to the intended design.

P1—separation of structure and instance values. S must describe reusable source/target locators, transformations, formatting, and parameter types; a particular instance supplies values through V_i. The only values permitted in S are constants explicitly prescribed by R_d, not values copied from a calibration output. **Status: implementation-backed at the representation interface.** `compiler.compile_cad_job` accepts `spec`, `source_data`,

and `runtime_params` separately, while `value_layer.resolve_field_raw` evaluates the value layer at execution time; `regression.RegressionRunner._run_case` reuses the same spec with each case's inputs. This interface supports reuse, although successful regression—not the type shape alone—is what detects a semantically overfit constant.

P2—explainable and verifiable locators. Every source or target locator must expose what it resolved to and must be checkable against the corresponding artifact; an unresolved or ambiguous locator is evidence, not permission to guess. **Status: implementation-backed.** `source_locator.r` returns the resolved sheet, cell, selection, and match count, and `source_resolution.verify_again` checks that the locator reproduces the derivation sample. On the target side, `target_ref.BaseTargetA` and `describe` provide a common existence and explanation interface. `docx_target.DocxTargetAd` and `xlsx_target.XlsxTargetAdapter.inspect` retain labels, physical locations, positional flags, and ambiguous references, while their `apply` methods return every missed reference.

P3—bounded layout resilience and fail-loud behavior. Semantic identities should survive admissible layout changes, while a change outside the declared tolerance must stop the run rather than trigger a guessed write. This includes the zero-LLM execution requirement: an execution-time model must not relocate a target. **Status: implementation-backed for the fail-loud half; the resilience half is bounded and measured rather than general.** Source locators resolve by sheet/header/section semantics; `docx_target.DocxTargetAdapter` distinguishes stable content-control/bookmark identities from positional table cells; `xlsx_target.XlsxTarg` distinguishes stable defined names from positional coordinates; and both reject duplicate or missing references. The execution branch that previously undermined this property — agent repair of a rejected job inside the run — has been removed, as described in Section 3.1.2, so a

target that cannot be resolved now ends the run with a code instead of being relocated.

What is *not* claimed is the resilience half in general form. Layout tolerance is declared per locator mechanism, and a change outside a mechanism's declared tolerance is detected only insofar as the mechanism can see it. Two measured boundaries are carried forward to Section 3.1.4 rather than glossed here: an unused detail-row cell that no column of a repeat block addresses is not cleared by `unused_slots: "blank"`, so a template reused from a previous reporting period can carry a stale value into a fresh run; and the derivation-time semantic oracle compares only references whose text differs between the baseline and the expected document, so a specification that writes into a cell the expected document leaves blank is structurally invisible to it.

P4—diffability, versioning, and lineage. The IR and its evidence must support reviewable changes, immutable publication, and regression over a template lineage. **Status: implementation-backed for the currently produced handle-backed field-map shape.** `models.DwgSpec` carries a lineage-scoped unique version and template fingerprint; `drift.spec_field_drift` reports added, removed, and attribute-level field changes; `models.DwgProductionVersion.save` prevents mutation of the published spec/hash identity; and `regression.RegressionRunner.run` persists results for all active cases of the lineage. The drift function reads the drawing-flavoured spelling of a target rather than the format-neutral list, which was recorded here as a prospective blind spot; Section efsec:design:drift reports the measurement that retired that concern and the narrower gap it found instead.

P5—independently auditable rule coverage. S must enumerate which rules and fields it claims to cover, and the denominator must come from the rule artifact independently of the model being scored. **Status: implementation-backed since the coverage-v2 path replaced**

agent self-enumeration. The original implementation — `rule_coverage.merge_inventory` over an agent-supplied `rule_inventory`, unioned across prior derives — did not satisfy P5 and is retained only to read historical specifications. A rule the agent never enumerated was absent from the denominator, so "22 of 22 covered" and "22 of 40 covered" were the same observation.

The active path externalizes the denominator instead. A dataset that carries a rule document must also carry a frozen, versioned rule-applicability manifest; `rule_coverage.parse_rule_manif` verifies that the manifest names the exact SHA-256 of the rule-document bytes and that its declared document roles match the dataset's, and `derivation_service.DerivationService._loa` refuses the derive with `RULE_MANIFEST_MISSING` before any paid model call, so a run cannot discover after the fact that its release denominator is unauditable. The agent supplies a *disposition* for each externally assigned rule identifier and nothing else: `rule_coverage.score_manifest` discards unknown identifiers, refuses duplicate dispositions, and refuses an assignment whose document roles disagree with the manifest, so the scored agent can neither add, remove, rename, nor move a rule between roles. The resulting partition into `covered`, `unresolved`, `rejected`, and `unmentioned` is mutually exclusive and sums to the frozen applicable set, and `gates.canonical_unresolved` turns every non-`covered` disposition into a publish gate rather than partial credit. Section 4.6 develops what this measures and what it still does not.

3.1.4 Measured Boundaries of the Specification IR

The five properties above state what the IR must do. This subsection states what it demonstrably does not, because a boundary that is only discovered by a reader running the system is indistinguishable from a defect the author did not notice. Each item below is either pinned by a

test or was measured on a real artifact; where an alternative shape was considered and rejected, the reason is given, since "we did not think of it" and "we rejected it" are different claims.

B1—`exclude` is a row-content predicate. A range's `exclude` clause inspects the content of a data row. It does not inherit meaning from a preceding section header, so a sheet whose rows alternate between "current month" and "cumulative" groups under one header cannot be partitioned by `exclude`. The rejected alternative was a domain flag such as `exclude_cumulative`; it was refused because it names one agency's form inside a carrier-neutral IR. The general shape adopted instead is a group projection (`records`: a fixed `group_size` with named row roles and per-column `select_offset`), which describes the same layout without naming the domain.

B2—stacked records assume a fixed group size. `records` models "g physical rows form one logical record" for a constant g. A record whose row count varies with its own data is outside the IR. Both the read side and the write side are implemented, and a specification produced by the agent for the XLSX evaluation template declares `records` with `group_size: 2` and named row roles, so the primitive is exercised by a main evaluation dataset rather than by fixtures alone.

B3—the verification basis for the collection primitives is $n = 1$. `artifact_collection`, `records`, and the equality merge of Section 3.5.2 each have exactly one motivating real workflow. Six independently sourced government workbooks were scanned for a second instance of a repeating multi-row record block and none was found; the nearest candidate was a one-off label merge, not a repeating record. The generalization threshold this thesis sets for a new primitive is therefore not met for these three, and they are presented as carrier-neutral shapes with a single validating workflow rather than as generally validated primitives. The risk this creates

is specific and worth naming: it does not appear as a failing test, because the tests were written from the same single example. It appears as a wrong result on a form nobody has tried.

B4—merge is equality-only and inner. Two resolved row sets can be joined on equal key columns. Range containment, interval overlap, and inequality correspondences are not expressible. The join is deliberately inner: supplying a default for an unmatched key would convert a missing record into a normal-looking number without anyone having authorized the default, which is precisely the silent-failure shape this thesis measures. A specification that needs the filled row must produce it on the source side, where the decision is visible. Merges chain, so three tables can be joined as two binary merges; the agent produced a chained pair unprompted.

B5—the aggregation axis is columns only. Aggregation sums named columns. The criterion is who decides the number of columns: if the template does, naming them is legitimate; if *this instance's data* does, the specification has become instance-specific, and the correct modelling is a long table plus a pivot at write time. A row axis is not offered because a wide table's column names would then be data values, which would disable the author-time static check that a formula only names columns the row set actually has.

B6—a repeat block clears only the cells its own columns address. When a block declares `unused_slots: "blank"`, the cells cleared in an unused slot are the cells that block's columns point at. A cell in the same slot that no column addresses is not cleared, because the IR knows a block's targets only through the strings its columns declare, and "every other cell on this row" is not something it can name. For a pristine template this is exactly right, and every instance of both evaluation datasets starts from a pristine template. For the common practice of reusing last period's completed report as this period's template, it means a stale value can sit on a row the system considers written. The behavior is pinned by a test rather than left to inference,

and the rejected workaround—declaring a column whose value is always blank—was refused because it puts layout knowledge into the value layer and makes the column count grow with the template's width.

B7—one instance is one filled target template. A specification may declare several output documents, but their roles must be distinct; a run that would produce M documents of the same role is refused. Relaxing this is a separate decision and is not claimed here.

3.1.5 Design Principle: Fail Loud, Not Silent

A *silent failure* is an output that is complete, well formed, correctly formatted, and wrong. It is the failure mode this thesis is organized around, and every design decision in this chapter that looks unnecessarily strict is a consequence of it.

The asymmetry is economic rather than aesthetic. A loud failure costs a rerun: somebody sees an error code, reads the reason, and fixes an input or a rule. A silent failure costs whatever the wrong number causes downstream, and it costs it *after* the document has been signed, filed, or paid against, because nothing in the delivery path distinguishes it from a correct document. The two are not two points on one scale of quality. A system that refuses too often is annoying and, crucially, *measurable* — its refusals are counted. A system that guesses is neither: its guesses are counted as successes. Optimizing a template-filling system for the number of fields it manages to produce therefore optimizes it in exactly the wrong direction, and the procedures in Chapter 4 are arranged to resist that pressure. The failure-mode taxonomy of Section 2.3 is the vocabulary this creed is stated over.

The same principle at three levels. The two earlier subsections state this creed twice without naming it, and it is worth collapsing them, because the three statements are not analogies —

they are the same rule applied to three different artifacts.

1. **The produced document.** This is the familiar level, and property P3 of Section 3.1.3 is its statement: a target that cannot be resolved ends the run with a code rather than being relocated by a model. The forbidden behavior is "locate approximately, then write".

2. **The specification.** A specification can be quietly wrong in ways no single execution reveals. The information boundary of Section 3.1.1 is the statement at this level: if the rule artifact R_d were permitted to carry target addresses, a derivation that merely copied them would be indistinguishable from a derivation that found them. The specification would look derived; nothing about it would be. The forbidden behavior is "supply the answer, then measure the search".

3. **The measurement.** A metric can be quietly wrong while every mechanism under it is correct. Property P5 is the statement at this level: while the coverage denominator was enumerated by the agent being scored, "22 of 22 covered" and "22 of 40 covered" were the same observation, and no test could fail. The forbidden behavior is "let the thing being measured choose the denominator".

Each level has its own way of being plausible while wrong, and therefore needs its own loud channel; a guarantee at one level does not propagate to the others. This is why the thesis reports semantic correctness, extent diagnostics, presentation, untouched regions and package integrity as separate numbers (Section 5.5.1) instead of one accuracy figure. An aggregate is a fourth place for a silent failure to hide.

What the principle forbids, concretely. The creed is not a slogan in this system; it is a set of refusals that each cost something. The source-side extent gates are the clearest examples, because each one refuses a value that would otherwise look entirely normal: `source_locator`

refuses an extent that selected zero rows (returning 0 for "sum of no rows" is indistinguishable from a genuine zero); refuses an extent that swallowed a subtotal row (the result is exactly twice a plausible number); and refuses a blank or non-numeric cell inside a numeric aggregate rather than coercing it to zero, which is the most common way a spreadsheet quietly underreports. The value layer applies the same rule one stage later: its aggregate helper raises rather than coercing, explicitly so that it cannot become a second, laxer gate behind the first. The equality merge of Section 3.5.2 is inner for the same reason (Section 3.1.4, B4): defaulting an unmatched key would convert a missing record into a normal-looking number that nobody authorized. A duplicated bookmark name or content-control tag identifies nothing and is recorded as ambiguous rather than resolved to whichever copy came first. And the one construct in the IR whose whole effect is to *relax* a publish gate — a declaration that a reference is deliberately left blank — is checked three ways at derivation time, and a declaration that fails any check is not attached to the specification at all (Section 3.3), because a declaration that both fails and inflates the coverage numerator is a silent failure of the measurement level while looking like a caught defect at the specification level.

The price, stated rather than hidden. Fail-loud has a bill and this thesis pays it in a visible place. Both of the best specifications produced for the two main evaluation carriers are, as measured, blocked from publication: on the XLSX template the merged publish gate reports 13 blocking items and on the DOCX pair it reports 8. Neither number is a defect count; both are correctness-preserving refusals, and Section 4.5 shows that reading them as a defect count is itself an error, because the same repository has reported "N unresolved" from two different counters that disagree in both directions.

Status: implementation-backed as a set of named refusals; *not* a proof that no silent failure remains. The principle is realized as error codes and gates that can be enumerated — among them

```
LOCATOR_RANGE_EMPTY, LOCATOR_RANGE_CONTAMINATED,
SPEC_REPEAT_TARGET_OVERLAP,
SPEC_REPEAT_SLOT_ALIGNMENT_UNVERIFIED,
SPEC_FORMULA_INPUT_UNPRODUCED,
CAD_JOB_MANIFEST_REJECTED, CACHED_HANDLE_NOT_FOUND
```

and `benchmark/probe/guard_sweep.py` mechanizes the rule that adding a new refusal must not quietly retire an older one: it disables each guard in turn and requires that guard's own tests to fail, and a row that selects no tests is treated as invalid rather than as passing. What must not be inferred from any of that is that the produced documents contain no silent failures. The thesis has a measured counterexample from its own system: on the XLSX evaluation template the derived specification declared no column for the remarks field, five calibration and development instances were executed, and three of the resulting cells were scored `silent_failure` by the shared evaluator — output that was complete, plausible, and wrong. That case is discussed as a result rather than as an exception in Sections 4.7 and 5.5.1; here it fixes the strength of the claim. The system is built so that the *known* ways of being quietly wrong are loud, and it measures how many of them remain, but "fail loud" is a design discipline with an evidence trail, not a guarantee.

3.1.6 Scope and Applicability Conditions

Section 3.1.2 introduced the target problem class in the course of arguing for compile-once execution. This subsection states it as an admission test, since Chapter 1 refers to it and

Section 6.2 develops what happens when it fails. Three conditions must hold together; each is stated with the question one would actually ask before adopting the method.

C-I — Recurrence. The same template, or a lineage of revisions of it, is instantiated many times. *Operational test:* can one name the expected number of instances per template revision, and is it large enough that a one-off derivation cost is amortized? The derivation is the expensive, stochastic step; nothing about the method is cheap at $N = 1$. Recurrence is what makes the trade in the first place, and it is why Section 5.4 reports a break-even N rather than a cost ratio.

C-II — Deterministic derivability. Every output value is a function of structured source data and the rules, computable without judgement. *Operational test:* for each field, can two competent people, given the same sources and the same rule document, be expected to produce the same character string? If a field requires professional discretion — a risk rating, a recommendation, a summary in someone's words — it fails this condition, and the correct response is to leave it outside the method rather than to widen the method. The evaluation enforces this rather than assuming it: the path that lets a person supply a value at execution time exists in the code and is disabled at derivation time (Section 3.7), so a derived specification cannot classify a field it found hard as one a human will fill in.

C-III — Explicit operational rules. The mapping is written down as rules over the source data, not held as tacit practice. *Operational test:* does an artifact exist which states, for each field, where the value comes from and what is done to it — and would a new employee be handed that artifact? Note what this condition does *not* require. It does not require the rules to be machine-readable, complete, or even correct: both evaluation datasets use prose rule documents, and building one of them found three rule statements that disagreed with the real form, one of

which pointed at a location that does not exist. It does, separately, forbid the rules from carrying write coordinates, which is the constraint of Section 3.1.1 and is a condition on the *experiment* rather than on the task.

The three conditions are independent. A recurring task with tacit rules fails C-III and is a documentation problem before it is an automation problem. A well-documented, deterministic task performed once fails C-I and should simply be done by hand or by an agent. A recurring, well-documented task containing one discretionary field fails C-II for that field only, and the honest decomposition is to carve the field out — which this thesis does not claim to have measured, because it disabled the mechanism that would allow it.

Status: normative, with the selection bias disclosed. These conditions are definitional, not empirical: no experiment in this thesis varies them. Both evaluation datasets were chosen *because* they satisfy all three, which means the thesis measures the method inside its declared scope and says nothing about how it degrades near the boundary — for instance, how a specification behaves when a rule document is twenty per cent incomplete, or when one field in twenty needs judgement. The bias direction is favourable to the method and is recorded as such in Section 5.10. The consequences of each condition failing are discussed in Section 6.2; this subsection deliberately gives only the definitions.

3.2 Domain-Dependent and Domain-Independent Layers

Whether this work is a tool for one file format or a method that happens to have been implemented for three depends on where the format-specific code stops. This section states that seam, because it is a claim about the contribution rather than an implementation detail: everything on the neutral side is shared by all three carriers as the same code, not as three

parallel implementations that agree today.

Table 1: Where format knowledge is allowed to live.

Carrier-dependent	**What it must answer**
Structured extraction	What is in this file, as a dump: the drawing dumper through the CAD gateway; `DocxTargetAdapter.inspect` and `XlsxTargetAdapter.inspect` in process.
Target location, write-back, and the baseline-to-expected diff	The operations of the `target_ref.TargetAdapter` protocol, plus `addressable` and `rebind`.
Package preservation	`xlsx_package`: which bytes must survive a write (Section 3.6).
Carrier-neutral	**Why it can be**
The IR and its normalizer	`spec_ir` speaks of documents, fields, extents, repeat blocks and blank targets; none of those words names a format.
Source locators and row sets	`source_locator`, `row_set`: a source workbook is read the same way whatever the output template is.
The value layer	`value_layer`: formulas, rounding, formatting.
The compiler	`compiler.compile_cad_job` produces a carrier-neutral write plan that `document_production` or the CAD gateway then executes.
Gates, oracles, coverage	`spec_check`, `gates`, `rule_coverage`.
Lineage, fingerprint, drift, regression	`models`, `fingerprint`, `drift`, `regression`.

Two properties of this seam are load-bearing. First, the carrier-dependent operations are the *only* ones a new format must supply: `target_ref.BaseTargetAdapter` implements `diff`, `verify` and `describe` once over a single per-format primitive, so a format states what its addressable targets are and inherits the rest. Writing those three per format is how a `.docx` diff quietly starts reporting something subtly different from a drawing diff, which no reviewer can see because the two are never displayed side by side. Second, the neutral side is neutral in its *vocabulary* as well as its code: a specification's write target is normalized to `{kind, ref}`, so "which places does this specification write" is one function rather than one per carrier.

The seam is not the module namespace, and confusing the two has cost something. The Python package is still called `features.dwg` and the compiler entry point is still `compile_cad_job`; both names are historical, from the drawing-only prototype, and renaming them would couple a cosmetic change to models, migrations, API paths, frontend types and the CAD case-study contracts. The names are therefore left alone — but the *messages* were not neutral, and that was a real defect rather than an aesthetic one. Until it was corrected, a failed specification validation returned "DWG spec validation failed" for every carrier, so an Excel run whose specification was rejected told its reviewer that something about AutoCAD had failed; two XLSX specifications were in fact rejected that way. The sibling drawing error codes were audited at the same time and are correctly confined to the drawing branch. The general lesson is recorded in Section 4.2: a correct check whose message names the wrong repair costs a reviewer as much as a wrong check, and no test covers the wording.

Status: implementation-backed, with one measured way the seam has been crossed accidentally. The table above describes the current module graph, and the `TargetAdapter` protocol makes it checkable rather than aspirational. What it does not guarantee is that every *path* through the neutral code has been exercised by every carrier. The measured counterexample is Section 4.1's: the single-document branch of the derivation service did not carry repeat blocks at all, because the only dataset that had ever used them produced two documents and therefore took the other branch — a correct agent reply was persisted with the detail table missing and no error raised. Neutrality of the code is therefore claimed; uniform coverage of the code by both carriers is not, and Section 5.10 records it as a threat.

3.3 Specification IR

The IR is the artifact the whole method exists to produce: the reviewable, versionable, executable object that a derivation emits once and every later instance consumes. This section describes what it addresses; Section 3.1.4 has already stated what it demonstrably cannot.

Scope: one instance, not one file. A specification describes the whole output set of one instance. Its root is a list of documents, each with a role, a template reference, a list of fields, a list of repeat blocks, and a list of blank-target declarations; declarations more than one document reads — named extents — live in a shared section beside them. The unit is the instance because that is the unit of the task: a customs shipment produces a commercial invoice *and* a packing list from one set of rules, and deriving them separately is neither the same unit of work as the agent baseline it is compared against nor safe, since two independently declared extents can disagree about which source rows are detail rows and nothing at execution time would notice.

An earlier generation of the IR had a single field list at the root, and twenty-one stored specifications still use it. They are not re-derived. `spec_ir.spec_documents` reads either generation as the same list, and every consumer goes through it. This normalizer is the point of the module rather than a convenience: a consumer that reads the root field list directly finds it empty on a new-shape specification and reports a successful run that wrote nothing, which is this thesis's definition of a silent failure.

A field. A field carries a name, a value source (`source`, `computed`, `constant`, `carry_over`, or the disabled `user_input`), the source key or semantic `source_locator` that supplies its value, an optional transform and transform type, an optional render format, an optional aggregate over a named extent, a human-language

description carrying no meaning to the compiler, and a target. Names are global across documents: one shared formula namespace is what makes a cross-document rule expressible at all — "the two documents' totals agree" is exactly that shape — and the cost, that two documents reusing an obvious name collapse into one, is reported as a warning rather than guessed at.

A repeat block. A block declares a detail table once and is expanded deterministically into the template's fixed slots, so the size of a specification stops depending on how many detail lines the calibration instance happened to have. It names the extent it iterates, the number of slots the template offers, where the first slot starts, how many physical rows form one logical record together with the named role of each, what happens to a slot the instance has no data for, and its columns; each column names its target through a template string containing a slot placeholder, and reads either a source column of the extent or a per-row expression over it. The one policy for an unused slot is to blank it, and deliberately only one: leaving a slot alone keeps the previous batch's row visible in the produced document.

Blank targets. A document may declare references it deliberately does not write, so that a rule whose obligation is negative — "this cell must stay blank" — can be implemented rather than left unresolved. This is the only construct in the IR that relaxes a publish gate, so a declaration must pass three derivation-time checks: every reference must be addressable on the template, must be blank in the expected artifact, and must be written by nothing else in the specification. A declaration failing any of them is reported together with the check that failed and is *not* attached to the document, because attaching it would let a false claim into the coverage numerator even while a finding blocked the publish.

The IR is also a published contract, and the contract does not currently hold. A JSON Schema for the specification ships in the contracts package, and it is a real constraint rather than documentation: several of its object definitions close `additionalProperties`, so a key the schema does not know is illegal rather than ignored. That strictness has already caught the failure mode this repository pays for most often — an allow-list that silently drops a key — but in the opposite direction. When the repeat-block definition was checked against a real specification for the first time, it turned out to have been forbidding, for two design revisions, the row-set columns the implementation had been emitting all along, because no test had ever validated a real specification against it.

That check was extended for repeat columns but not for the corpus. Validating every stored specification against the published schema now gives 24 of 35 valid and 11 invalid, from eight distinct causes. Three are legacy specifications predating a required key and one is a legacy row shape; the remaining seven are live divergences between what the backend emits and what the contract permits, including the aggregate key introduced for detail-table totals on a field, provenance and label keys the derivation attaches to repeat blocks and their columns, and a null render format where the schema requires a string. The three most recent specifications for the XLSX evaluation template are valid; a fresh derivation of the DOCX pair would not be, because the field-level aggregate key it needs is still not in the contract.

Status: implementation-backed for the IR and its normalizer; the published contract is measured and does not hold over the stored corpus. `spec_ir` is a pure module with a normalizer both generations pass through, and its shape checks are the same functions the write side and the read side use, so a specification cannot be validated by one vocabulary and executed by another. The contract gap is a real defect and is handled under the first of this thesis's three routes for such a gap — fix the implementation, with a recorded entry naming what it

blocks — rather than by weakening the claim: the correct statement today is that the IR *has* a published contract which 24 of 35 stored specifications satisfy, not that the contract is an enforced invariant. The acceptance criterion recorded with that entry is the one this measurement makes obvious: a check that validates the stored corpus, not another hand-written fixture, since a fixture is written from the same understanding as the code it is meant to contradict.

3.4 Structured Dump as Grounding

Nothing in the derivation opens a template. The agent is shown a *dump*: a structured inventory of what a template contains and where it can be written, produced by deterministic code before any model call. This is the grounding step, and three properties of it matter for the thesis.

One shape for three carriers. A drawing dump is produced by the CAD gateway; a `.docx` or `.xlsx` dump is produced in process, since neither has a gateway, a worker, or a cold start to time out on. Both are then reshaped into the same envelope — a list of entities, each with an opaque reference, its current text, and its kind — with the richer per-target detail (caption, positional flag, physical location) riding alongside. This is not a fudge: because the target reference is already defined as a format-neutral identity, reshaping here means the baseline-to-expected diff, the field-map derivation, the evidence record, the compiler, the verifier and the publish gates all work on a Word template without a second implementation of any of them. Two things are deliberately kept *outside* the entity list. A document's full text lines ride beside it, because a regression case comparing against an approved document must see text the specification never mentions, while a paragraph is not a write target and putting it in the entity list would make target verification accept it as a place a field may write. A workbook's sheet titles are recorded explicitly, because a blank cell is not a target, so a sheet whose write points

are all still empty would otherwise be invisible in the dump.

Content addressing, and the version of the dumper. Dumps are cached and keyed by tenant, the artifact's content hash, the inspection level, the target kind and the scope, so re-deriving over unchanged bytes performs no extraction, and the cache disposition per template is reported with the result rather than hidden. The key carries a dump schema version for a reason that is easy to miss: a dump is a pure function of the file's bytes *and of the dumper*. Content addressing covers only the first, so changing what the extractor emits without touching any file leaves every key matching and serves stale-shaped dumps indefinitely; bumping the constant invalidates them by construction. The target kind is in the key for the analogous reason — a `.docx` dump and a drawing dump are different shapes of answer to the same question, and one must never be served for the other.

Honest disclosure: the carrier libraries lose things. An extractor is only as faithful as the library under it, and this one has measured losses. Across the 29 real `.xlsx` files in the repository, a load/save round trip through the spreadsheet library silently drops the calculation chain and the shared-string table (both harmless, since Excel rebuilds them), the printer settings and the sheet relationships that reference them, cell comments together with their legacy drawing part, chart colour and style parts, and — only when an optional imaging dependency is absent — every embedded image. That last one was found the expensive way: the dependency was present in the development environment, declared nowhere, and absent from a container build, so images disappeared from written workbooks and every run reported success.

Two consequences are worth separating, because they land in different places. As a *carrier* limitation it belongs here: what a spreadsheet library can round-trip bounds what any method built on it can promise, and it is not a limitation of compile-once execution. As a *measure-*

ment problem it belongs in Section 3.6, because for a period the system under test lost exactly the parts that the benchmark's package-integrity channel injects as its fault, which made that channel unable to tell a correct run from a deliberately broken one.

Status: implementation-backed, with the extractor's coverage stated as a bound. The cache key, the version constant and the two carriers' in-process extractors are all in the production path and are exercised by the derivations reported in Chapter 5. What is *not* claimed is that a dump enumerates everything a template can express. A spreadsheet dump offers only cells that carry content — measured on the real monthly-report template, 27 of them, while the entire data area the specification writes ships blank — and that measurement is the reason target verification and target addressability are two different questions in Section 3.6 rather than one lenient one.

3.5 Locator Mechanisms

A locator is the durable, re-executable rule for finding something by meaning rather than by address. The distinction is the method's centre of gravity: an absolute cell reference such as `B153` is correct exactly until a row is inserted, a column is moved, or a line of descriptive text is added above the table, and when it stops being correct it does not stop producing a value.

3.5.1 Semantic Locators

A single-cell source locator names the artifact role it reads, the sheet by semantic name with accepted aliases, and the cell by the intersection of a row filter and a column header. Four properties make it a locator rather than a search.

The header row is discovered, not assumed. Leading descriptive text above a table does not

defeat the locator, because the row that carries the declared column headers is found rather than fixed at the first row.

Identification is by header text only. No geometric alignment of x and y positions is used, so moving a column leaves the locator valid. This is a deliberate narrowing: alignment is the spreadsheet analogue of matching a drawing entity by its coordinates, which the derivation prompt bans outright.

Stacked blocks are scoped before anything else happens. Many real sheets stack several blocks vertically, each introduced by a standalone label row and each repeating the same column headers. A section clause confines header discovery, row filtering and selection to one block. Without it a column header resolves against the first block's header row and a maximum sweeps every block, so the locator returns the right number from the wrong block — and only breaks on the next customer's workbook.

Sheet names are resolved across a set of files, and collisions fail loudly. A rule document says "the shipment detail sheet", not "the third upload", so sheets from all source artifacts share one namespace. Two customers' workbooks both containing a sheet called `Sheet1` is entirely ordinary; that used to resolve to whichever file came later, with a warning nobody read, so the value came from the wrong workbook while every file involved looked perfectly normal. A locator that resolves to a colliding title now fails, a locator that names its artifact searches only that file, and a collision nothing looks up is not an error.

Resolution returns the resolved sheet, cell, selection and match count, so what a locator did is reviewable rather than inferred; zero matches, an ambiguous match, and a missing sheet or column each return a structured error rather than a guess.

Two shapes of block, not one. The header-driven table above is one of the two shapes real forms use. The other is a key/value block — a label in one column and its value in the next,

continuing down the sheet — which has no header row to discover and where the row's own label *is* the key. It is supported as a distinct declaration rather than as a special case of the first, and the distinction reaches surprisingly deep: the "is this row blank" test that bounds an extent must consider the whole row for a header-driven table, because the noise that contaminates an extent lands wherever the form had space, and must consider a single column for a key/value block, because the label to its left belongs to a different field and a whole-row test reads past the end of the block.

Status: implementation-backed on the source side; the label-proximity target mechanism is unexercised on both main carriers. Everything above is exercised by both evaluation datasets. The related mechanism that locates a *write* target by proximity to its caption is not: it is built on the non-agent derivation path, both main datasets take the agent path, and across the 33 specifications examined all 13 label-geometry matches belong to three drawing specifications from an earlier phase of the project. Sections 4.7 (D3) states this as a boundary of what author-time verification has seen; the consequence here is a restriction on what this chapter may claim, and it is a narrow one: labeled-field target location is a case-study capability demonstrated on drawings, not a demonstrated capability on DOCX or XLSX.

3.5.2 Range Locators and Source-Side Aggregation

Almost every business form contains a subtotal, a maximum, or a count. Expressing one requires a locator that denotes not a cell but *a run of rows reduced to a value*, and that single change turns out to generate most of the interesting design in this thesis. The history is worth telling as a sequence rather than as a finished mechanism, because each step was forced by an observation rather than anticipated, and the sequence is itself the argument: **the expressiveness boundary of this IR was pushed outward by datasets, not designed in advance.**

Step 1 — the extent, and why it is the hard part. A range locator adds to a single-cell locator a declaration of which rows it covers and how they are reduced. A single-cell locator asks "which cell?"; a range also asks "from which row to which row" — does it include the subtotal line, the notes row, the blank separator, and is it still right after next month's insert? An extent that is one row wrong produces a number that is plausible and wrong, which is precisely the failure this thesis exists to make impossible to miss.

Three commitments follow. The extent is always *declared*, never inferred: there is deliberately no whole-column form, and the one open-ended option must be explicitly marked unbounded so that it surfaces in review rather than at the next revision. The reduction refuses the three shapes that hide an error — an extent that selected nothing, an extent containing a row whose text marks it as a total, and a blank or non-numeric cell inside a numeric aggregate. And the resolution records the rows it included *and* the rows it excluded with a reason, so a reviewer who can see the extent can see the off-by-one row that a correct-looking total would otherwise hide.

Step 2 — a run of rows reduced to a scalar is not enough. The first specification derived for the DOCX pair touched table rows one, two and three of a template with fourteen detail slots, because the calibration instance happened to have three detail lines and the agent wrote them out one by one. That is not compile-once; it is compile-per-instance, and the cause was not the prompt but the IR: a range could be collapsed into a scalar, and there was no way to say "the nth slot takes the nth row of the extent". The repair was a row projection — a repeat block over a named extent, expanded into the template's fixed slots — together with a specification scope of one instance's whole output set, so that two documents reading the same detail table cannot disagree about which rows it contains.

Step 3 — a repeat column that can only read one source column is not enough. The next derivation produced a well-formed repeat block whose columns could each read a source column and nothing else, so a quantity that is "per carton $\times$ cartons", a unit price that must be looked up in a product master, and every total over the resulting values were all inexpressible. The agent diagnosed this itself in the specification it returned. The repair moved the evaluation stage: an extent is resolved *once, into a table* of per-row namespaces; joins add columns from other artifacts; computed columns evaluate over that table in declaration order; and a field's aggregate reduces one of its columns.

The decision worth recording is where the computed columns live. They belong to the *extent*, not to the repeat block that displays it. Two of the DOCX rules are "the packing list's quantity equals the invoice's quantity on the same line" and "both documents have the same number of lines, and line n is the same part". If each document's repeat computed its own quantity, those rules would be satisfied only by two formulas happening to agree, and a divergence would appear as two plausible numbers on two documents that nothing compares. One column, computed once and read by both, makes them agree structurally — and it is also what allows a total to name `extent.column` at all, since a value computed inside a repeat exists in no table anything could aggregate.

Step 4 — a slot placeholder with neither an origin nor a stride. The XLSX evaluation template put the data area at rows 6 to 19, two physical rows per logical record. The slot placeholder substituted the slot ordinal directly, so slots one, two and seven addressed rows one, two and seven: any sheet whose data area does not begin at the first row was unaddressable. The repair gives a block a *layout* — a slot origin, a group size, and named row roles with per-column offsets — so that a slot's coordinate is computed rather than assumed. This defect survived a long time for a reason that is itself a finding: the only dataset that had ever used repeat blocks

was the DOCX pair, whose detail slots begin at table row one by coincidence, so the XLSX repeat write path had never run on a real layout and no test could have been red.

Step 5 — two resolved tables that must be joined by key. The monthly report pairs a project register against per-project audit results, and expressing that required an equality merge between two already-resolved row sets. Before it existed, the only way to pair two tables was to point two repeat blocks at the same slots — an *implicit positional merge* whose correctness depends on both sides having the same ordering *and* the same membership. Removing the second of those, by changing one project number so that one side has a member the other does not, moved every audit figure down one row while both declared-count gates passed and every written cell remained a legal integer. That is why the merge exists and why a separate gate now refuses two blocks that share a slot range without a declared join.

Status: implementation-backed at every step, each with the observation that forced it. Steps 1 through 5 are all on the production path and all exercised by at least one evaluation dataset; the resolved-table representation is used by the derivation-time oracle as well as by execution, which is what gives a detail table any semantic check at all (Section 4.3). What this section deliberately does not claim is generality for the newest primitives: the boundaries of the resulting representation — equality-only inner joins, a column-only aggregation axis, a fixed group size, and a verification basis of one real workflow each for the collection, record and merge primitives — are stated in Section 3.1.4 as B2 to B5 rather than repeated here.

3.5.3 Failure Modes and What Each Mechanism Blocks

Table 2 maps the failure modes of Section 2.3 onto the mechanisms of this chapter. Its purpose is to show that the mechanisms were selected against a taxonomy rather than accumulated,

and equally to show where the taxonomy is covered by something other than a locator — three of the six rows are not locator problems at all, which is itself the reason the locator mechanisms are not the whole method.

Table 2: Failure modes and the mechanism that is supposed to make each one loud.

Failure mode	**Mechanism**	**Residual**
Wrong value, right place	Author-time semantic oracle over the expected artifact; the resolved row set makes detail cells comparable	Only references whose text *changed* between baseline and expected are compared (Section 4.7, D4)
Locator drift after a template revision	Structural fingerprint plus the publish gate; stable identities (defined name, bookmark, content control) preferred and positional references flagged	A change beyond the last content-bearing cell moves no fingerprint (Section 3.8.3)
Coverage shortfall — a field no rule reaches, so the template default survives	Externally frozen rule-applicability manifest; every non-covered disposition is a publish gate	Coverage is a derivation diagnosis, not an execution correctness measure (Section 4.6)
Format and unit error	Render formats in the IR; decimal half-up rounding (Section 3.7); presentation scored as its own channel	Only the rounding function has been audited for decimal semantics
Negative-obligation violation — what should be blank is not	Implied blanks derived from row roles, written rather than merely checked; explicit blank-target declarations	A target template that appears in no column is outside the derivation (Section 3.1.4, B6)
Extent error — an aggregate covering one row too many or too few	Declared extents with the three refusals of Section 3.5.2; recorded included and excluded rows; per-range attribution in the evaluator	A hypothesis is only distinguishable where the dataset made it so (Section 5.5.4)

Where the locators currently stand, measured. An earlier draft of this section carried a locator failure rate from the drawing phase — thirty-one mapped rules of which twenty-one had no confirmable write location. That number is obsolete and it flattered nothing: it predates every mechanism in Section 3.5.2 and both main carriers. The current measurements are the rule dispositions of the two best derivations. On the DOCX pair, 33 of 34 applicable rules are covered, 0 rejected, 1 unresolved. On the XLSX template, 23 of 27 are covered, 0 rejected, 4

unresolved. The four unresolved on the XLSX template are informative because none of them is a locator failure: three are negative obligations — rules whose content is "this must stay blank" — whose declarations the derivation could not verify against the expected artifact, and the fourth is a remarks column for which the specification declared no column anywhere, which the execution probe later confirmed produces silent failures.

Status: implementation-backed for the mapping, normative for its completeness. Every mechanism named in Table 2 exists and is exercised, and every residual in the third column is measured and cross-referenced rather than asserted. What is normative is the claim that the six rows *exhaust* the failure modes: the taxonomy comes from the literature and from this project's own incidents, and a mode nobody has hit is indistinguishable in this table from a mode that cannot occur. The bias direction is favourable to the method, and Section 5.10 records it.

3.6 Target Reference and Write-Back

A target reference answers "where does this field write?". All three carriers offer a stable, author-declared identity and a positional fallback, and the design decision is to support both while making the difference visible rather than to support only the safe one:

- **Drawing:** an entity handle, an opaque identity the drawing itself maintains across edits.
- **Word:** a content-control tag or a bookmark name — stable, author-declared, document-global — and, positionally, a table cell addressed by part, table, row and column, or an underscore blank addressed by paragraph and index.
- **Excel:** a defined name — the exact analogue of a handle, since Excel maintains it across inserts — and, positionally, a sheet and coordinate.

Refusing the positional forms would mean refusing most real templates. Measured on a genuine forwarder commercial invoice, the three stable forms addressed the line-item table and nothing else: all eleven header fields produced zero targets, plainly visible in the text dump and unwritable. So positional references are supported and flagged `positional`, and a reviewer can see which targets will not survive a row insert instead of discovering it at the next revision. The captioned blank is flagged `volatile` in addition, because filling one blank renumbers the others on the same line.

Verification and addressability are two questions. "Does something exist at this reference?" and "can something be written at this reference?" coincide for a handle, a bookmark and even an empty Word table cell, which exists in the XML regardless. They do not coincide for a spreadsheet cell, which denotes a position that exists whenever its sheet does and whose normal state in a template is blank. Collapsing them made a *correct* seven-slot repeat block on the real monthly-report template report 41 of its 42 cells as unwritable, and the only way to satisfy the check would have been to point the block at cells that already had content. The two questions are therefore two methods, with addressability defaulting to verification for every format that does not override it. What the spreadsheet override still catches is the failure that matters at block scale — a wrong sheet name, after which every slot lands nowhere and no value-level check sees it. What it cannot catch is a wrong column or a shifted row; neither can the writer, which writes them happily, so no check on this dump ever could, and that gap is closed by comparing against the references the expected document changed.

Writing back a volatile reference requires a second reading. Filling the blanks in a Word template destroys the references that named them: the survivors renumber, so the produced document answers the original reference confidently with a different field. Discarding the produced

document's own volatile references and re-reading them by paragraph coordinates is therefore not an optimization but a correctness requirement — and the coordinate chosen is the paragraph ordinal plus that paragraph's literal text, because those are the only two things both writers preserve. A produced document whose paragraph numbering differs from the template's leaves every volatile reference unresolved rather than resolvable-but-shifted, which keeps the failure loud.

Locating a target is the first of three layers, not the whole problem. It is tempting to treat "the reference resolves" as the criterion for a correct write. It is the first of three, and this project has been wrong at each of the other two.

1. **Can the coordinate be found?** This is what target location and addressability answer. Section 4.7 (D1, D2) records what the author-time side of this layer can and cannot see — in particular that an expected output supplies candidate coordinates that introspection alone does not, and that the Word blank locator is a pilot-supported volatile mechanism rather than a general one.

2. **Does the write preserve the package?** A correct value in a correct cell is still a failed run if the file that comes out has lost the parts that make it the form. Writing through a spreadsheet library's save re-serializes the library's own model of a workbook, so every part the library has no model for is simply absent from the output. On the real monthly-report template that is the drawing part holding the ruled boxes and stamps, and the printer-settings part holding the one-page layout a filed form must keep. The repair replaces the writer with a package editor: open the original archive, rewrite only the one worksheet part that has to change, and copy every other part through byte for byte, while the spreadsheet library stays in place answering only *where* to write, since merged-range anchors,

defined names, data validations and conflicting references all need a spreadsheet model. Measured on five calibration and development instances, the package-integrity channel moved from $0/5$ to $5/5$, and after one write the only changed part is the worksheet.

3. **Does the produced document actually say what was planned?** The compiler knows exactly which references it will write and with what text, so the produced artifact is checked against that plan — every planned reference holds the planned text, and nothing unplanned changed. The last clause is the one that matters, and it cannot be established from the output alone; it needs the baseline to diff against.

The three layers are independent, and the middle one is worth a sentence of its own because it is the only one whose failure was invisible to the evaluation rather than merely undetected: the benchmark's package-integrity channel injects its fault by performing exactly one spreadsheet-library round trip, so while the system under test wrote through the same library, a correct run and a deliberately broken one produced the same class of file and neither $5/5$ nor $0/5$ was evidence about anything. The acceptance criterion for the repair was therefore written as "the two are distinguishable" — the clean output must score full marks and a deliberately round-tripped copy of the same output must score zero — rather than as "the broken one scores lower".

Status: implementation-backed, with two residuals named. The adapter protocol, the addressability split, the volatile rebinding and the package editor are all in the production path, and each has a negative control that makes its own guard fail when disabled. Two residuals: the package editor covers XLSX only, and the Word layout parts have not been enumerated the same way, which matters because the DOCX pair is the only DOCX evaluation dataset (Section 5.10). And the package editor does not decide what *may* be written — preserving every

part does not make a value written into the wrong column any less legal.

3.7 Deterministic Compiler and Zero-LLM Runtime

The compiler turns a published specification plus one instance's inputs into an executable write plan and a content hash of it, and returns failures *before* anything is submitted. Its two design commitments are that errors surface at compile time rather than during execution, and that evaluation is restricted enough to be auditable.

A restricted expression language, not a sandboxed one. Formulas are parsed to an abstract syntax tree and evaluated by an interpreter that accepts a closed list of node types: literals of string, number and boolean; names; the arithmetic and comparison operators; a conditional expression; and calls to a fixed function table (`min`, `max`, `round`, `abs`, `sum`, `avg`, `count`, `count_nonblank`). Everything else — attribute access, subscripting, comprehensions, keyword arguments, assignment — reaches a final clause that raises with the offending node type named. A name the context does not provide is an error rather than an empty value, which is what makes it possible for the derivation's trial compile to catch a formula referencing something nothing produces; that check is the single most productive author-time gate in Chapter 4, and it is only possible because the language is small enough to have no fallback.

Rounding: the rule says one thing and the language does another. Every rule document in both datasets says "round half up". The implementation initially used the host language's built-in rounding, which is round-half-to-even, and the diagnosis "switch the rounding mode" was wrong — following it would not have fixed the defect. The measured case is a volumetric weight, $35 \times 25 \times 20 \times 3 \div 10^6$, which the rule, the spreadsheet, and a human all read as the decimal literal 0.0525 and round to 0.053. As a binary double its exact value is $0.05249999\ldots$,

which is not a tie at all, so half-up rounding applied to that value still returns 0.052. The value has to be pulled back to the shortest decimal literal that round-trips to the same double before the digit the rule talks about exists to be rounded; only then does half-up quantization give the answer the rule specifies. The repair therefore has two steps, not one, and it disposes of the whole family of cases usually written off as needing arbitrary-precision arithmetic throughout.

The reason this earns a paragraph in a thesis rather than a line in a changelog is that it is a perfect specimen of the primary risk. Both 0.052 and 0.053 are entirely normal-looking figures — right unit, right precision, right format — and every gate in the system passed them. Sweeping the DOCX dataset's 79 detail rows found three disagreements, all volumetric weight, and one of them is in an *evaluation* instance. Left unfixed, a rounding-mode defect in the host language would have been recorded as an error of the method and would have contaminated the main result table.

Human input at execution time: available, and deliberately disabled. A field may in principle take its value from a person at execution time. This does not call a model and therefore does not violate the zero-LLM claim, and it is a reasonable capability for a real system, so the code remains. It is disabled at derivation time for a reason of *evaluation validity*: derivation defines such a field as the residual — text that changed and that no source value explains — so allowing it would let the arm being scored decide which fields it is not responsible for. That is the same error shape as letting the scored agent enumerate its own coverage denominator, and it is asymmetric across arms, since a per-instance agent has no such escape hatch and simply has to produce a value. The closure is placed at derivation rather than at execution on purpose: closing it at execution broke 34 tests, because the twenty-one stored drawing specifications are built on runtime parameters, and that would have confused "this thesis's evaluation does not credit human input" with "this system does not support human input". Reopening it is an evaluation

change before it is a code change: the input manifest would have to declare per field which values are human-supplied, and all three arms would have to receive the identical strings.

The zero-LLM red line. Execution must contain no model fallback: a target that cannot be located must fail and be reported, never be guessed. That requirement, the removal of the execution-time repair branch, and the negative-control test that pins it are stated in Section 3.1.2 and are not restated here. What belongs in this section is the consequence for the compiler: because there is no fallback, every failure mode has to be expressible as a compile-time refusal with a code, which is why the error vocabulary of Section 3.1.5 is as large as it is. A smaller vocabulary would not mean a simpler system; it would mean a system that continues past conditions it cannot name.

Status: implementation-backed. The compiler is a pure function of specification, source data, runtime parameters and current template values, and the canonical job it emits is content-hashed, so identical inputs are recognisably the same compiled work. The restricted evaluator, the two-step rounding and the derivation-time closure of human input are each pinned by tests, and the rounding repair carries a negative control on the specific measured case. Two limits are worth stating: the decimal audit covers the rounding function only, and no other numeric function has been checked for decimal semantics; and content-hash equality establishes that two runs compiled the same work, not that the produced files are byte-identical, which is a separate check (Section 5.4.1).

3.8 Lifecycle Management

A specification that is correct once is a demonstration. A specification that stays correct across template revisions, is reviewable when it changes, and can be traced from a delivered

file back to the exact program that produced it is an artifact an engineering process can accept. This section describes the four mechanisms that make the difference, and it is where the thesis's governance argument against per-instance agents is grounded — not in a claim about regulation, which this thesis does not make, but in traceability, reproducibility and change control, which are properties one can point at.

3.8.1 Lineage and Versioning

Three identities are tracked and related.

A *template lineage* is the family of revisions of one customer template, and it carries the current structural fingerprint of each document role it covers. A *specification* belongs to a lineage, carries a version number unique within it, and records the fingerprint of each template role it was derived against — one per role, because a specification produces an instance rather than a file, so the packing list's template drifting invalidates it even though the invoice's targets are untouched. A *production version* freezes one approved specification together with the compiled artifact and its content hash; those four fields are immutable after creation and the model refuses to save a change to them, so only activation and rollback may move. An *execution run* then references its production version, the exact compiled artifact it submitted, and a unique correlation identifier.

The result is a three-way trace in both directions: from a delivered file to the run, from the run to the compiled program and its hash, and from the hash to the specification version and the template fingerprints it assumed. The published program's identity hash is computed over the field map, the rule projections and the runtime parameter schema, deliberately excluding runtime values, so "the same program" and "the same inputs" remain separate questions.

3.8.2 Regression Suite

Regression cases belong to the lineage rather than to a specification version, so a new derivation is judged against the same cases as the old one. Each case pins its own input template, source artifacts, source-data override and runtime parameters, and optionally an approved expected output. A run compiles the specification against the case's inputs, executes it, and checks the result twice: against the compiler's *plan* — every planned reference holds the planned text and nothing unplanned changed — and, when the case has one, against the approved expected document as a stricter full-text comparison.

The plan check is the load-bearing half, and it is available for every case including those with no approved output. It is also what pairs with the fingerprint: the fingerprint deliberately excludes text, so a caption renamed in place with the same structure does not move it, and the regression's expected comparison is what catches exactly that. The two compose — structural drift is caught by the fingerprint, semantic drift by regression — and a failing regression blocks publication.

3.8.3 Drift Detection

The fingerprint is a stable hash over a template's *structure*: which references exist and, for each, its type, its layer, and its attribute tag where it has one. Text is deliberately excluded, because text is the payload the pipeline exists to rewrite — folding it into the identity would churn the fingerprint on every production run and invalidate every specification for no structural reason — and because no reliable rule distinguishes a static caption from a value by content alone. Geometry and dump ordering are excluded for the same class of reason.

Measured on both main carriers, because it had only been tested on synthetic drawing dumps. The fingerprint's unit tests feed it hand-built entity lists in the drawing shape, so whether it discriminates on a real document was an assumption. Measuring it on the two evaluation templates gives Table 3. The controls matter as much as the positives: a library round trip with no edit at all must not move the fingerprint, or the mechanism would report drift every time anything opened the file.

Table 3: Structural fingerprint on the two evaluation templates.

Perturbation	**XLSX (27 refs)**	**DOCX (92 refs)**
Library round trip, no edit (control)	unchanged	unchanged
Caption text edited in place	unchanged	unchanged
Row / paragraph inserted inside the form	**changed**	**changed**
Row appended past the last content-bearing cell	unchanged	—

The last row is the honest one. On a spreadsheet the fingerprint is a hash over the cells that carry content, because a blank cell is not a target; a template edit that displaces none of them is therefore invisible to it. On the monthly-report template that means everything below row 25. This is a genuine boundary rather than a bug: the alternative — hashing the declared sheet dimension — would move the fingerprint whenever a cell that once held a value was cleared, since the dimension stays inflated long after the value is gone.

A skeleton-level claim this measurement retired. An earlier note in this chapter's outline asserted that drift detection "goes blind the moment a specification uses generic target references", on the grounds that the diff function reads the drawing-flavoured spelling of a target and not the neutral list. Measured across all 35 stored specifications — 585 fields, 582 target references — the diff function sees 582 of 582, and *no* stored specification uses the generic spelling. The published contract, moreover, closes the target object to additional properties, so a specification using it would be contract-invalid: the branch is unreachable for any specifica-

tion the contract admits. The claim was therefore not so much wrong as untested and pointed at the wrong risk, and the accurate statement is narrower and duller: drift compares fields and repeat blocks, both of which it covers completely, and the neutral spelling is a capability of the reader rather than a shape anything currently produces.

What the same audit *did* find is one unit the diff does not compare at all: a document's blank-target declarations. Two specification versions that differ only in whether a reference is claimed as deliberately blank produce an empty structural diff, so a reviewer reading the change sees nothing. The consequence is bounded — such a declaration changes no write, and dropping one sends its rule back to blocking the publish through a different gate — so this is recorded as a stated boundary with an entry naming the rejected alternative, rather than as an implementation task ahead of the writing this chapter blocks.

3.8.4 Auditability

Publication is where every mechanism in this chapter is cashed in, and its structure encodes one distinction: which refusals a human may take responsibility for, and which nobody may.

Not overridable: a specification with no fields; a specification failing schema validation; a specification whose declared document roles do not match its dataset's; a specification that did not pass its derivation-time trial compile; and a specification for a multi-document lineage in which some template role has never been fingerprinted. The reasoning differs per item and is worth separating. "This does not compile" is not a judgement call — every run of it fails at the first step, so publishing can only produce a production version that never executes. "Nobody ever fingerprinted this template" is refused because it is indistinguishable in the data from "no template drifted", and publishing on the second reading means publishing with drift detection switched off for that document.

Overridable, with a price: the merged unresolved set, a stale template fingerprint, a failing regression run, and a missing review. A forced publish requires a written reason, and the reason, the operator, the exact list of gates waived, and the timestamp are recorded permanently on the production version and surfaced as a standing warning. The design intent is that overriding is possible — a process that cannot be overridden is worked around — but never quiet.

The merged unresolved set, and the two counters that disagreed. Everything unresolved is merged into one canonical set: items the agent could not resolve, changes in the expected document nothing accounts for, unresolved rule targets, source values whose locator failed or disagreed with the derivation sample, and specification fields carrying no verified target. One property of that merge is worth stating here because it is a general lesson about evidence rather than a detail. The cells a repeat block writes are routed into the "unexplained changes" list on purpose, because that is where the derivation-time oracle reads expected text from and it is the only way a detail table gets a semantic check at all. The publish gate reads the same list to answer a different question — "did anything account for this change?" — and for those cells the answer is yes, the block owns the address. Until that was separated, a *correct* specification carried dozens of spurious blocking items, and no test turned red because what changed was a count. The ownership set is recomputed from the field map rather than read from a tag in the evidence, so a later version that drops the block sends the same cell back to blocking.

The measured state of the two best specifications is 13 blocking items on the XLSX template and 8 on the DOCX pair, against 13 and 14 respectively in the agent's own unresolved list. The two counters therefore agree on one specification and differ by six on the other, and before the repeat-cell separation above they differed the other way, by 34 and 19. Every narrative of "N unresolved" in this project's records counted the agent's list; the merged set is the one that decides publication. Section 4.5 draws the methodological rule: any unresolved count must be

decomposed before it is interpreted, because “the gate is too strict” and “the gate is doing its job” call for opposite repairs.

Status: implementation-backed, with the governance claim scoped. Immutability of a published version is enforced at the model, the gate ordering and the forceable set are one pure function testable in isolation, the force audit is persisted, and the fingerprint behaviour in Table 3 is measured on the real evaluation templates rather than on fixtures. The scoped part is the argument this section supports: the claim is that this architecture supplies traceability, reproducibility and change control, and that a per-instance agent supplies none of the three because there is no versioned program to trace, no compiled artifact to hash, and no diff to review. The claim is *not* that any standard or regulation requires deterministic output; no such source is cited, and Section 2.4 states the requirement in those three words instead.

3.9 Expressiveness Boundary

This chapter has stated boundaries in three different places, which is a hazard rather than thoroughness: a reader who meets a limitation twice in different words cannot tell whether it is one limitation or two. This section therefore does two things — it says where each kind of boundary lives, and it states the one boundary that belongs to none of the other lists.

Division of labour. Section 3.1.4 (B1–B7) states what the *representation* can express: row-content predicates, fixed group sizes, equality-only inner joins, a column-only aggregation axis, what a repeat block clears, and the $n = 1$ verification basis of the newest primitives. Section 4.7 (D1–D6) states what *author-time verification* can see: which coordinates introspection alone yields, which locators are pilot-supported, which mechanism has never executed on a main carrier, which cells the semantic oracle compares, and what the shared evaluator is not. The two

are different questions and must not be merged — "a merge does not do outer joins" and "the oracle cannot see a cell that is blank on both sides" are not the same kind of statement, and a combined list would invite a reader to treat them as interchangeable. The present section holds what is neither: boundaries of *execution and lifecycle*, which is to say things the IR can express and the derivation can verify but the running system still cannot promise.

E1 — variable-length write-side detail. Reading is unbounded in the number of rows; writing is not. An instance with seven detail lines does not cause seven rows to be inserted into the template. The asymmetry is deliberate and the argument is blast radius: the read side is confined to two pure modules that take bytes and return values, while inserting rows on the write side reaches the target adapters (reflowing formats, merged ranges, formula references), the comparison logic (two documents with different row counts), the blanking logic, the structural fingerprint, and drift — and, worse, it shifts every positional reference, which means deliberately breaking the thing the system itself flags as fragile. The accepted alternative shape is a template with a fixed number of detail slots of which an instance uses some, the rest of which must be cleared.

That alternative was recorded as a limitation, and it is worth stating plainly that it is no longer only that. The clause "the unused slots must be cleared" turns a negative obligation into a first-class construct, and both halves of that are now implemented: the blanks implied by a record's row roles are *written* rather than merely checked — checking alone would make compliance a property of the template's pristine state rather than of the specification — and a document may additionally declare references it deliberately does not write, subject to the three checks of Section 3.3. So the residual boundary is narrower than the original statement: what remains unexpressible is a template whose row count is a function of the instance, not the negative obligation that the fixed-slot alternative brings with it.

E2 — fields requiring cross-document reasoning rather than lookup. A value that must be inferred from prose in another document, rather than read from a structured source and transformed, is outside the method by construction — this is condition C-II of Section 3.1.6 viewed from the implementation side rather than from the admission side. Nothing here degrades gracefully into it: there is no execution-time model to fall back on, which is exactly the point of the zero-LLM claim.

E3 — carrier fidelity bounds what any of this can promise. Two limits established earlier in this chapter are properties of the file formats and their libraries rather than of the method, and they bound its guarantees anyway: a spreadsheet dump enumerates only cells that carry content (Section 3.4), and a structural fingerprint therefore cannot see a change beyond the last such cell (Section 3.8.3). The package-preserving writer closes the corresponding *write*-side limit for XLSX and does not close it for DOCX, whose layout parts have not been enumerated (Section 3.6).

Status: E1 and E3 implementation-backed as boundaries; E2 definitional. E1's asymmetry is a design decision with the coupling argument recorded, and the negative-obligation half of its alternative shape is implemented and measured rather than promised; E3's two halves are each a measurement reported above. E2 is a restatement of an admission condition and carries no evidence, by construction. All three feed Section 6.3, which develops them as future work rather than inventing new ones.

Chapter 4. Methodology

Chapter 3 described the specification S and the deterministic execution that consumes it. This chapter describes the procedure that produces S: how a stochastic model proposal is turned into an artifact a deterministic layer has interrogated, what each interrogation can and cannot see, and where a human decision is required rather than optional. The organizing constraint throughout is the thesis's primary risk measure. A derivation procedure that maximizes the number of fields produced is easy; the procedure below is instead arranged so that every way a specification can be wrong is either detected before publication or is explicitly named as undetectable.

4.1 State Machine

Derivation is a fixed sequence of transitions in which a single stochastic step is surrounded by deterministic admission checks. Writing it as a state machine rather than as "call the agent and validate the result" is not presentation: the order of the transitions is itself a result, because three of them exist only because an earlier ordering wasted a paid model call.

Let Φ denote the transition sequence implemented by `derivation_service.DerivationServi`
Given $T^{(n)}_{\mathrm{der},d}$ it produces a persisted draft specification and its evidence, $(S, \mathcal{E}_S)$. The transitions are:

1. **Ground.** Each declared output template is inspected into a structured dump. Dumps are content-addressed and cached, so re-deriving over unchanged bytes does not re-extract

them; the cache disposition per template is reported with the result rather than hidden.

2. **Admit the denominator.** If the dataset carries a rule document it must also carry a frozen rule-applicability manifest. `DerivationService._load_rule_manifest` verifies the manifest against the exact bytes of the rule document and against the dataset's declared document roles, and fails the derive on any mismatch.

3. **Resolve the carrier.** The target kind is computed from the output artifact before the agent is addressed, because the agent is instructed differently for a drawing than for a document, and a `.docx` derive framed as a drawing task is told to call tools that cannot exist for it.

4. **Warm the drawing gateway,** for the DWG carrier only. Probing a CAD gateway for a `.docx` would fail against a perfectly healthy installation.

5. **Propose.** The agent receives the dumps, the staged original source and rule artifacts, the externally assigned rule identifiers, the field-name hints from prior non-degraded specifications of the same dataset, and the declared document roles. This is the only stochastic transition and the only paid one.

6. **Admit the proposal's addressing.** A reply naming a document role the dataset has no template for is refused with `AGENT_DOCUMENT_ROLE_UNKNOWN`. Silently dropping those fields would leave a specification that is merely short, while rule coverage still counted them as covered.

7. **Bind.** The per-document field map, repeat blocks, and shared extents are assembled, and target references the agent did not supply are recorded as unresolved with a cause that distinguishes an infrastructure outage from a rule that was not understood.

8. **Verify the source side.** Every declared source locator is re-resolved against the derivation's own sample artifacts and must reproduce the value the derivation used; failures and disagreements are recorded, not assumed away.

9. **Resolve the detail tables.** Named extents are resolved into row sets against the calibration instance's own workbooks, and their values are folded into the sample data. This costs no model call—it is the same deterministic pre-pass that execution runs—and it is what gives the semantic oracle of Section 4.3 anything to compare inside a repeat block.

10. **Gate structurally** (Section 4.2).

11. **Gate semantically** (Section 4.3).

12. **Repair, bounded** (Section 4.4).

13. **Make the evidence persistable.** The assembled evidence is checked for encodability before the write.

14. **Persist a draft.** Never a published specification, and never nothing.

Two properties of this ordering are load-bearing.

Admission before expenditure. Transitions 2, 3, and 6 all reject inputs for which the agent could nonetheless have produced a plausible answer. They are placed where they are because the alternative was measured: a dataset assembled without a declared output role was rejected in roughly half a second at zero model cost by `RULE_MANIFEST_ROLE_MISMATCH`, whereas the same defect discovered after transition 5 would have discarded the entire paid run. The general rule this encodes is that a check whose inputs are all available before the model call belongs before the model call, even when placing it later would be simpler to write.

Derivation fails to publish, not to produce. No transition after the agent's reply can cause the derive to return nothing. A specification that will not compile is still persisted, with the reason recorded, because a broken draft a reviewer can read is worth more than an error message. What the procedure must not do is let such a draft look healthy, and that is a separate mechanism: everything that survives unrepaired is written into `evidence.spec_validation`, and `gates.canonical_unresolved` turns it into a publish gate (Section 4.5).

Status: implementation-backed, with one boundary that was learned the expensive way. Transition 13 exists because the guarantee above was once violated by the evidence record itself. Evidence is assembled from a dozen sources and written to a JSON column as the last act of a derive; a spreadsheet cell read as a native date-time value is not JSON-encodable, and the exception it raised inside the database write destroyed a completed paid run, leaving not even a broken draft. The repair is deliberately narrow: `derivation_service._persistable_evidence` attempts the encode and returns the evidence byte-for-byte unchanged when it succeeds, converting only on failure and recording in the evidence itself that a conversion happened. An unconditional stringify would have hidden the underlying defect, which is that something is putting live objects into a record. The honest limit is that this guarantees persistence against encode failures that have been *observed*; it does not prove the evidence assembly has no other way to fail.

4.2 Structural Gates

The structural gate is the question "does this specification run?", asked of the deterministic layer rather than of the model that wrote it. Before this gate existed, derivation persisted whatever the agent returned; a specification declaring a lookup transform with no table to look in was

stored as an ordinary draft and discovered much later by a human running an offline checker. The agent was not being dishonest. Nothing had asked.

`spec_check.check_spec` returns every deterministic defect visible without the drawing gateway, as reviewable items carrying a code, the JSON path or field name, and a detail string. It has four parts, in a deliberate order.

Schema and shape. The IR validator rejects malformed documents, fields, extents, repeat blocks, row sets, merges, and record declarations. The IR shape validator alone accounts for 65 distinct extttSPEC_* codes out of roughly 200 prefixed error codes in the feature, and those codes are the vocabulary that the repair loop and the reviewer both read.

Policy. Runtime human input is disabled at derivation time, not merely at execution time, with `SPEC_USER_INPUT_DISABLED`. A field whose value comes from a person at run time is a residually defined field: it lets the party being scored decide which values it is not responsible for. Rejecting it at derivation rather than at execution also matters for the reviewer's diagnosis, because the execution-time symptom is "a runtime parameter is missing", which names the symptom and not the cause.

Source binding. `SPEC_SOURCE_FIELD_UNPRODUCED` reports a field whose declared source key nothing in the specification produces. It runs after the schema check, because a malformed field has no trustworthy source declaration to inspect, and before the trial compile, because the trial compile is precisely what hides this defect.

Trial compile. The specification is compiled against the derivation's own sample data twice: once with every declared runtime parameter supplied and once with only the required ones, which are the two branches an optional override takes. This is the important half. The worst real defect it caught was a formula referencing a name that nothing provided

— perfectly valid syntax, and invisible to any schema check.

The gate is also defended against itself. A compile that raises rather than returns an error is caught and reported as `SPEC_COMPILE_CRASHED` rather than allowed to propagate, for the reason given in Section 4.1: an exception here occurs after the agent has been paid and before anything is persisted, so a single malformed rule would discard an entire run. The behavior is pinned by `test_the_trial_compile_survives_an_exception_it_cannot_anticipate`

4.2.1 What a Structural Gate Catches That an Instance Run Cannot

The argument for spending effort here is that these checks run before any instance is processed and before any evaluation budget is spent. One concrete case makes the argument sharper than the general claim does. Two repeat blocks may write into the same set of slots, each reading its own row set, so that slot n takes its values from the nth row of two different tables. That is an *implicit positional join*, and its correctness depends on two conditions holding simultaneously by coincidence: that both row sets are ordered identically, and that both contain the same members. Remove the second condition — give one source file a project number the register does not contain, without changing the number of files — and every value shifts by one row. Row-set resolution succeeds. Both independent declared-quantity gates pass, because no quantity changed. Every written cell is a well-formed integer. Nothing downstream can see it.

The gate added for this shape, `SPEC_REPEAT_SLOT_ALIGNMENT_UNVERIFIED`, refuses a pair of blocks that share a slot origin and group size while reading different, unrelated row sets. Two things about it are worth recording as method rather than as engineering. First, the failure it addresses was present in a specification that had already passed every other check. Second, its collateral damage was measured rather than assumed: replayed against all thirty-four previously derived specifications it fires on exactly two block pairs, both true positives, and

zero times on the five specifications of the other evaluation dataset.

Status: implementation-backed, with one known ordering defect. When the schema layer reports an error, `check_spec` returns before running the trial compile, and the source-binding check is likewise skipped. The short circuit has a reason — a structurally broken specification has no reliable source declarations, and a compile error stacked on a schema error only restates it in less specific language — but the cost is that one derivation reveals one layer of problems, and seeing the second layer costs another paid run. This is a real defect against the cost argument of RQ3 rather than against correctness; it is recorded as an open item, and the intended repair is to run the source-binding check over the fields that are structurally sound instead of skipping it for the whole specification.

A methodological rule this section is an instance of. Adding a gate changes what the gates in front of it can still observe, and no test turns red when an older protection becomes redundant. Two protections in this system were measured to have become vacuous in their own test suites exactly this way. The discipline adopted in response is that a new gate is accepted only after each older gate it sits in front of has been disabled in turn and its own test confirmed to fail. This is the negative-control requirement of Section 4.3 applied to guards instead of to measurements.

4.3 Two Complementary Oracles

The structural gate establishes that a specification runs. Whether it is *right* is a different and much harder question, and it is asked twice, by two oracles that see different things. Neither subsumes the other, and the argument that they are complementary is the central empirical claim of this chapter.

4.3.1 The Author-Time Expected-Text Oracle

A derivation dataset that supplies an expected output already states what each target reference should read, and the derivation's own sample data is what produced it. The comparison is therefore available at author time, before any instance is processed and at no model cost: compile the specification against the sample, and compare the planned text at each reference with the expected document's text at that reference. `spec_check._expected_text_warnings` implements this.

What makes it worth a section is the class of defect that nothing else sees. Consider a detail-line quantity that the rule document defines as units per carton *times* the number of cartons, written instead as a sum. The two operands are small integers; the sum is 43 where the product is 120, and the amount that follows from it is 550.40 where the correct figure is $1{,}536.00$. Both results are entirely normal-looking numbers: right currency, right decimal places, right thousands separator. The formula is valid, so the structural gate passes it. Before this oracle existed, the first indication would have been an evaluation run, after the money was spent, and only if the scorer itself had no correlated bug. The example is pinned by a test rather than merely described.

Three exclusions are necessary, and under-excluding is the dangerous direction, because every exclusion missed becomes an accusation against a rule that is in fact correct. A field fed by a runtime parameter is excluded, because the trial compile supplies a placeholder rather than the real input. A two-cell row is excluded, because without current values it collapses into a single string. Anything computing from a field that had to be dropped as invalid is excluded, because it is computing from a value the complete specification would have supplied differently.

Two extensions of this oracle are results rather than details.

The key must include the document role. Under multi-document specifications a target reference identifies a cell in *one* file and is not unique across a specification, so the same reference string exists in both output documents. Keyed by reference alone, the oracle told the agent that the packing list's field was writing the wrong value; the repair loop obligingly rewrote that field to read the invoice's key; and the specification was persisted reporting `ok: True` while planning a letter-of-credit number into a shipping-port field. **The system's own correctness check manufactured a silent failure.** This is the strongest single argument in this thesis for the position that a verification mechanism must be verified with the same suspicion as the thing it verifies. The oracle was not absent, and it was not obviously broken. It was half-keyed, and a half-keyed oracle is worse than no oracle, because it actively drives repairs in the wrong direction.

The detail table was structurally invisible until three separate blockers were removed. The detail table is the largest and most computed part of a specification — seven columns across fourteen slots plus six totals, in the DOCX evaluation dataset — and it was the only part with no author-time semantic check at all. Two blockers were known in advance: the comparable-reference map read only document fields, and a repeat cell had no author-time value, because its value came from resolving a locator against a workbook that the trial compile does not have. Resolving extents into row sets at transition 9 removes the second. The third was not anticipated: every cell of a detail table is written by a repeat block, so no *field* ever claims it, so it never appears in the per-field evidence — it lands among the unexplained changes instead. With both known blockers fixed and the third still present, the measurement was that all 140 repeat cells had a real author-time value and the oracle could compare *zero* of them, because the expected side of the comparison was empty. The number of mechanisms that were individually correct while their composition measured nothing is the point; this is the same shape as the

vacuous-guard problem of Section 4.2.

4.3.2 The Case-Level Oracle

The second oracle compiles the specification against a regression case's own inputs and compares the resulting write plan against what that case says each target must read, yielding four verdicts per reference: `match`, `MISMATCH`, `MISSING`, and `OVERWRITE`. `OVERWRITE` is not redundant with `MISMATCH`: it is the verdict for rewriting a cell the case says must retain its baseline text, which is the negative obligation that a purely value-oriented comparison cannot express. References the specification writes that the case does not model at all are reported separately as out-of-scope rather than as failures, so a specification is not penalized for a cell the annotation does not cover.

The two oracles are complementary because they fail independently. The author-time oracle sees only the single calibration instance the derivation was given, so a rule that is correct for that instance and wrong in general — a constant copied from the sample, an extent that happens to end where the sample's data ends — passes it and is caught only by the case-level oracle. The case-level oracle in turn needs annotated cases, which cost human effort per instance, and it runs after the specification exists, so it cannot prevent a defect from being proposed; it can only refuse it afterwards. Their combination is what allows the expensive oracle to be spent on the defects the cheap one structurally cannot see.

4.3.3 Why Not an LLM Judge

The obvious alternative to both is to ask a model whether the produced document is correct. This thesis refuses it, and the reason is specific to the primary risk measure rather than to a general preference for determinism.

Silent failure is an output that is complete, well-formed, plausible, and wrong. The errors a generating model makes and the errors a judging model makes are drawn from correlated distributions: they share a training distribution, a tokenizer's numeric habits, and — if the judge belongs to the same family — a prior over what a filled form "should" look like. A judge is therefore least reliable exactly where the measurement matters most, because the outputs it is most likely to accept are the plausible-and-wrong ones. A judge would also make the measurement irreproducible, which forecloses the run-to-run identity this thesis claims for execution. The same reasoning governs the evaluator of Chapter 5, which is deterministic for this reason and not merely for convenience.

Status: implementation-backed, with three boundaries that must be stated.

1. **The author-time oracle sees only references whose text differs between the baseline and the expected document.** The comparable map is assembled from per-field evidence and unexplained changes, and both collect changed cells only. The consequence is exact, and belongs in the limitations rather than in a footnote: a specification that writes a value into a cell the expected document leaves *blank* is structurally invisible to this oracle — which is precisely the negative-obligation shape that the detail tables of both evaluation datasets depend on. What currently prevents the failure is that a misaligned block will almost always also collide with some cell whose content did change, and that is an accident of layout rather than a property of the design. The intended repair is a separate warning for "the specification claims a target the expected document leaves empty", not an enlargement of the comparison map, because pouring every blank cell of a template into evidence would both inflate it and lose the distinction between deliberately blank and out of scope.

2. **Expected artifacts and the ground truth used to score them share an origin.** Both are produced by a hand-written per-dataset generator from the rule document, which satisfies the red line that expected outputs must not be produced by the system under test. It does not make them independent of *each other*: a misread rule produces a consistently wrong pair, and the manual recomputation that would catch it has been completed for one calibration instance of the XLSX dataset only. This is disclosed as a threat rather than repaired, and the direction of the resulting bias is toward agreement, not toward detecting error.

3. **Any measurement of zero requires a negative control.** A reported "no mismatches" is not evidence unless a deliberate corruption has been shown to make the count nonzero. This is not a stylistic preference: verification probes in this project returned a serene zero while measuring nothing on four separate occasions — from a wrong key name, a wrong field name, a comparison performed over an empty set after an upstream failure, and a control that changed every addition including a string concatenation, so that row-set resolution failed entirely and the result was $0/0$ rather than a disagreement. A control must also distinguish doing the work from not doing it: an ordering test whose expected order coincided with the arrival order stayed green with the sort disabled, and had therefore measured nothing since the day it was written.

4.4 Case-Repair Loop

Both oracles produce failures that a model could plausibly correct. Allowing it to is a research risk and not merely an engineering one: the agent is being shown the expected answers, and "make case 3 pass" is trivially satisfied by a rule that special-cases case 3's numbers and destroys case 2. Such a specification is *worse* than the broken one it replaced, because it now

passes the tests that were supposed to catch it. The repair mechanism is therefore defined by its refusals.

4.4.1 Two Loops, Deliberately Distinct

The derivation-time loop, `DerivationService._verified_spec`, runs inside transition 12 and closes over the structural gate and the author-time oracle. The case-level loop, `case_check.repair_against_cases`, closes over the case-level oracle. They share their anti-overfitting rules and differ in what a regression means, because the case-level loop has a per-case signal and the derivation-time loop does not.

4.4.2 Four Invariants

I1 — a repair may only touch the value layer. The keys an agent proposal may change are enumerated in one place: which formula, which lookup table, which conditional cases, which format, which constant, which source key, which aggregate. Target references, source locators, runtime parameters, and the set of fields are re-imposed from the original afterwards, so a "repair" can never quietly become a rewrite of where a value comes from or where it goes. Anything the agent attempted outside that set is reported rather than applied, which keeps a rejected proposal auditable instead of invisible.

I2 — a deterministic check owns the verdict; the agent only proposes. Every candidate is re-checked in full before it is taken, and a candidate that increases the error count is rejected outright: a repair that breaks more than it fixes is not a repair. Errors dominate warnings in that comparison, because trading a specification that compiles for one that agrees with the expected document but does not run is never the better deal. A candidate that ties on both is nonetheless accepted, since the agent may have rewritten a rule into a clearer form the checks cannot

distinguish.

I3 — at the case level, regression is judged by *which* references fail, not how many. A count is not enough: swapping a failure on one reference for a failure on another scores identically, so a repair could move the damage elsewhere and still be accepted as long as some other case improved. The guard therefore compares failure sets, and a case that already failed to compile is exempt, because otherwise the moment it started compiling would be scored as a regression.

I4 — the loop is bounded and hands off to a human. The derivation-time loop makes at most three attempts and stops early when the agent proposes nothing admissible, since an identical second round would only cost another call. The case-level loop stops after two consecutive rounds without a net gain. The rationale for stopping rather than continuing is substantive: at that point the ambiguous thing is usually the rule document itself, and no amount of additional model effort resolves an ambiguity in a requirement.

4.4.3 A Refusal That Was Correct

One incident is worth recording because it inverts the usual reading of a failed repair. Across two rounds the agent declined to add a source aggregate that the reported errors clearly called for. The diagnosis was not agent laziness: the repair prompt states that only the value layer may be modified, that key was not on the permitted list, and the resolved row set was not included in the repair payload. **The agent's refusal was the correct behavior for the contract it had been given.** The defect was in the contract. This is why invariant I1's permitted set and the repair prompt are treated as a single artifact: a repair loop constrained by two documents that disagree will spend paid calls producing correct refusals, and the symptom is indistinguishable

from a model that cannot do the task.

Status: mixed, and the mixture must not be blurred. The derivation-time loop is implementation-backed and runs on every derive; the four invariants above are pinned by tests, including tests that a repair may not move a target reference, may not invent or drop fields, and may not be accepted when it worsens any single case. The case-level loop is implemented and tested as pure functions, and is reachable through an offline operator tool that runs a stored specification against its cases without the drawing gateway; **it has no caller in the automated derivation path or in the API.** Following this thesis's rule for a gap between claim and implementation, the explicit route is taken rather than the silent one: the case-level repair loop is presented as a designed and tested mechanism whose automated integration is future work, its absence is recorded as an open item, and Chapter 5 must not attribute any measured result to it.

4.5 Human in the Loop: Evidence Review

The economic claim behind compile-once is not that human effort disappears but that it changes units: from reviewing every produced document to reviewing one specification once. That claim is only honest if the review is actually possible, which imposes requirements both on what derivation records and on what publication refuses.

4.5.1 What Is Recorded

A derived specification persists an evidence record with fourteen top-level keys. On the most recent derivation of the XLSX evaluation dataset they include: the agent's own analysis (its unresolved items with causes, its field inventory, its reconciliations, its rule dispositions, and the observed tool-call health), the per-reference changed text and its count, per-field confidence,

field renames, locator verification, row-set resolution, rule coverage, the specification-validation report, unexplained changes, unresolved labels, residual source keys, and version drift against the previous specification of the same lineage. Two of these exist specifically to keep movement from reading as churn: drift reports added, removed, and attribute-level changes, and the rename report identifies fields that are the same field under a new name. This matters because field names drift between derivations — measured twice — so field identity for review purposes is established by target reference, never by name.

4.5.2 What Publication Refuses

`gates.canonical_unresolved` merges every unresolved signal into one set, de-duplicated by kind and reference: agent could-not-resolve items, unexplained changes in the expected document, unresolved labels, source locators that failed to resolve or disagreed with the sample, rule dispositions that are not `covered`, coverage-validation failures, and the specification-validation errors and warnings the repair loop could not clear. Any nonempty result blocks a normal publish. The design intent is stated in the module itself: nothing may be silently dropped into "ready to use".

Publication is gated on that set together with template-fingerprint staleness, regression results, and human review. A human may override the reviewable gates, but the override is not free: it requires a written reason, and exactly what was bypassed is recorded permanently on the published version. An empty or structurally invalid specification is not overridable at all.

Two design decisions inside this gate are worth stating, because both were mistakes first. *Causes are carried, not collapsed.* An unresolved field says nothing on its own about why it is unresolved, so a gateway outage and a misunderstood rule appear identical in the specification while being fixed in completely different places. The run's observed tool health is therefore

folded into each unresolved item's cause, and the review interface says so first and plainly, because a reviewer who read an outage as a coverage regression went looking for a rule that was never missing. *The headline number is* `covered`*, never* `accounted`*.* Accounted is the total minus the unmentioned, so it counts a rule the agent enumerated and then admitted it could not implement as though the work were done: fifteen rules implemented, sixteen reported unresolved, and zero unmentioned, once rendered as a green "34/34". Both halves of that display were wrong in the same direction — the flattering one — which is why the direction of an error, and not only its size, is treated as a reportable property throughout this thesis.

Status: implementation-backed as a mechanism; normative as an economic claim. The evidence record, the merged unresolved set, the forced-publish audit trail, and the review interface all exist and are exercised by tests. What is *not* measured is the quantity the economic argument needs: how long a review takes, how much of the evidence a reviewer actually uses, and whether a reviewer detects a defect the gates missed. No user study has been conducted. Chapter 5 therefore reports human cost only as the count and kind of items a reviewer is required to adjudicate, and the threats-to-validity discussion discloses that this understates human cost, since adjudicating an item is not free.

4.6 Coverage Is the Real Bottleneck

Once a specification's fields are individually accurate, the residual risk is dominated not by fields that are computed wrongly but by rules that no field implements at all. A rule the specification never addresses produces no field, no unresolved entry, and no gate: it is invisible by construction. Measuring that invisibility is therefore a methodological problem before it is an engineering one.

4.6.1 Why the First Denominator Did Not Work

The original implementation asked the agent to enumerate the rule document and counted its enumeration. This cannot answer the research question, for two independent reasons. A rule the agent never noticed is absent from the denominator, so "22 of 22 covered" and "22 of 40 covered" are the same observation; and omitting a rule *raises* the score, which points the incentive the wrong way. Worse, the visible figure was not comparable between runs: one derivation counted 39 mapped fields and the next 28, reading as a large regression, when nine of the 39 were filler the compiler now handles itself and the truth was a small improvement. Unioning inventories across derivations made omissions monotonic but left the evaluated agent choosing the items being counted.

4.6.2 The External Denominator

The active mechanism externalizes the denominator. A rule document must be accompanied by a frozen, versioned rule-applicability manifest that lists rule identifiers, their prose, and the document roles each applies to — and nothing else, in particular no target coordinates, which would make the manifest an answer key. The manifest is bound to the rule document by the SHA-256 of its exact bytes and to the dataset by its declared document roles, and a derive without it is refused before the agent runs.

The agent supplies a *disposition* for each externally assigned identifier and can do nothing else to the denominator: unknown identifiers are discarded, duplicate dispositions are refused, and an assignment whose document roles disagree with the manifest is refused. The resulting partition into `covered`, `unresolved`, `rejected`, and `unmentioned` is mutually exclusive and sums to the frozen applicable set. A claimed implementation is checked, not accepted: a `covered` disposition must name at least one field that really exists in the specification under

a role where the manifest says the rule applies, and otherwise becomes `unmentioned` plus a deterministic issue. Everything that is not `covered` blocks publication; there is no partial credit.

4.6.3 Metric Plumbing Is Methodology

The most consequential finding of this section is not the manifest. It is that a denominator, an allow-list, or an accepted spelling silently decides a reported number without failing anything, and that this failure mode is separate from — and harder to see than — a wrong mechanism. Three measured instances make the point.

1. **A denominator that admitted the wrong checks.** The scoring policy initially counted the "this slot must remain blank" obligations — 120 of them on one instance — inside the semantic denominator. An arm that submitted the blank template unchanged, writing nothing at all, therefore scored 66.7% on semantic correctness. After separating those obligations into their own channel it scores 0%. Every individual check was correct; what was wrong was which total they were added into.

2. **An accepted spelling that decided the headline.** Rule dispositions may name a field either bare or qualified by its owning block. One derivation used the qualified form throughout; the function computing the numerator accepted only bare names, so eleven correct assignments were reported as naming an unknown target, and coverage was displayed as 8 of 27 when the specification in fact covered 19. The specification and the assignments were both correct. A naming convention had decided the thesis's headline diagnostic number.

3. **Declared data that nothing read.** Each range in the benchmark manifests declares, for every wrong-extent hypothesis, the values that hypothesis would produce. The shared

evaluator did not read that field at all. Once connected, an output that occupies one extra detail slot while leaving all six totals byte-identical passes all 112 semantic checks and is nonetheless *named* as a specific wrong-extent hypothesis. The detection capability had been authored, and was simply not wired up.

The rule adopted from these is stated as a verification obligation rather than as advice: whenever a headline figure moves, confirm that the mechanism changed and not the bookkeeping.

4.6.4 What Coverage Does and Does Not Measure

Coverage is a derivation diagnostic. It states which rules the specification claims to implement, verified against an external list; it does not state that an implemented rule is correct, which is the oracles' question, and it must not be substituted for evaluation correctness.

A single coverage number for a dataset is also an incomplete description of the method, and the honest form is a distribution. Four independent derivations of the same XLSX dataset — with the prompt, the payload, and the input bytes identical — produced coverage of 15, 12, 19, and 23 out of 27 respectively, alongside structural differences of the same magnitude: one run declared three detail slots where the template has seven, another declared none of the cross-file collection its predecessor had used, and the number of hard structural errors ranged from four to zero. This variance is not noise around the method's performance; it *is* the method's variance, and the formulation of Chapter 3 predicts exactly where it lives. Because $\mathcal{A}$ is stochastic while $\mathcal{X}$ is deterministic, all of the run-to-run variance of this approach is concentrated in derivation and none of it is in execution. That is the substance of RQ2, and reporting a single derivation's coverage without the distribution would misstate it.

Status: implementation-backed, with the residual gaps named. The external denominator, the four-state partition with its checksum, the pre-agent admission of the manifest, the numera-

tor validation, and the publish gate are all implemented and tested. Three things remain outside the mechanism. First, the manifest is authored by a human from the rule document; binding it to the document's bytes prevents drift but does not prove the human segmented the prose correctly, and no independent segmentation of the rule documents has been performed. Second, the applicable-set hash proves which rules were counted, not that the set is the right one. Third, coverage says nothing about a rule that is covered *wrongly*; that separation is deliberate, and it is why this chapter needs both this section and Section 4.3.

4.7 Boundaries of Author-Time Verification

Section 3.1.4 listed what the IR cannot express. This section lists what the derivation procedure cannot see, which is a different set. Each item is stated in the form that makes it checkable: what is claimed, what is not, and which direction the resulting bias runs.

D1 — an expected output supplies target coordinates that introspection alone does not. When the dataset includes an expected document, the difference between baseline and expected locates the cells that must be written, including cells the baseline leaves blank. That is what makes blank-target binding work at derivation time. It does not follow that an unmarked blank paragraph or cell has a stable semantic anchor when only the template is available: without the expected document there is frequently nothing to distinguish one blank from its neighbours. Derivation-time capability and introspection-only capability are therefore separate claims, and only the first is demonstrated.

D2 — the DOCX paragraph-blank locator is a pilot-supported volatile locator. The mechanism that addresses a blank run inside a Word paragraph resolves against the literal textual context surrounding it in the template. It is supported for the DOCX evaluation dataset and is not

presented as general Word blank-target addressing; a template edit that changes the surrounding wording can invalidate it, which is why the write-back layer distinguishes stable content-control and bookmark identities from positional ones and fails loudly on the latter rather than guessing.

D3 — the label-proximity mechanism has never executed on either main evaluation carrier. Labeled-field resolution is constructed only on the non-agent derivation path, and both evaluation datasets derive through the agent. Of thirty-three stored specifications, the thirteen fields whose target was matched geometrically all belong to three DWG specifications from an earlier period; the DOCX and XLSX datasets contribute zero. This thesis therefore does not claim that labeled-field location works on DOCX or XLSX, and the measurement that recently tightened that mechanism was likewise taken entirely on DWG specifications and does not transfer.

D4 — the author-time oracle sees only references whose text *changed*, as established in Section 4.3. It compares text against text, so a target that is blank in the baseline and blank in the expected document produces no diff entry and is never compared. Two of the three failures that follow from this have since been closed, and the third has not; separating them matters, because the original single statement of D4 was used to justify a protection that turned out to be a coincidence.

The first failure — the specification writes a value into a reference the expected artifact keeps blank — is now detected by a distinct check rather than by the oracle. Derivation records the set of references at which each expected artifact holds text, and any planned non-empty value at a reference outside that set is reported. On a fixture whose repeat block is offset by one row, it names the two cells carrying a project name in a row required to stay blank; the test that pins that fixture previously recorded, in words, that the oracle said nothing about them. Two

silences in that check are load-bearing and are kept: a planned blank never warns, because an unused slot resolving to nothing is required behaviour and warning on it would accuse every specification whose calibration instance is shorter than the template; and a document for which nothing was recorded never warns, because absent evidence is not evidence of a blank.

The second failure — the specification cannot state that it deliberately writes nowhere — is now expressible, in two forms. Where a repeat block declares column positions for only some of its record's row roles, the uncovered positions are *derived* as implied blanks and written blank, with no new key in the representation: on the XLSX dataset this yields 21 cells that the block owns and blanks. Where the cells lie outside every block — a signature box, for instance — the document declares them, and the declaration is checked at derivation time against three conditions: every reference must be addressable on the template, must be blank in the expected artifact, and must not be written by anything else in the specification. A declaration failing any of the three is refused and does not reach the persisted specification, which keeps it out of the coverage numerator; this construct is the only one in the representation that *relaxes* a publish gate, and without those conditions an agent could dispose of any rule it failed to locate by swearing the cell stays blank.

The third failure remains open and is the residual meaning of D4: a reference that neither the expected artifact nor the specification writes, and that no rule names, is invisible. There is nothing at it to compare and no declaration pointing at it. The bias direction is unchanged and is towards accepting a specification: the oracle can only accuse, and a reference it cannot see is never accused.

D5 — an unpublishable specification is not by itself evidence of a failed derivation, and the count must be decomposed before it is read. Two separate findings sit under this heading, and both were closed as defects rather than being boundaries.

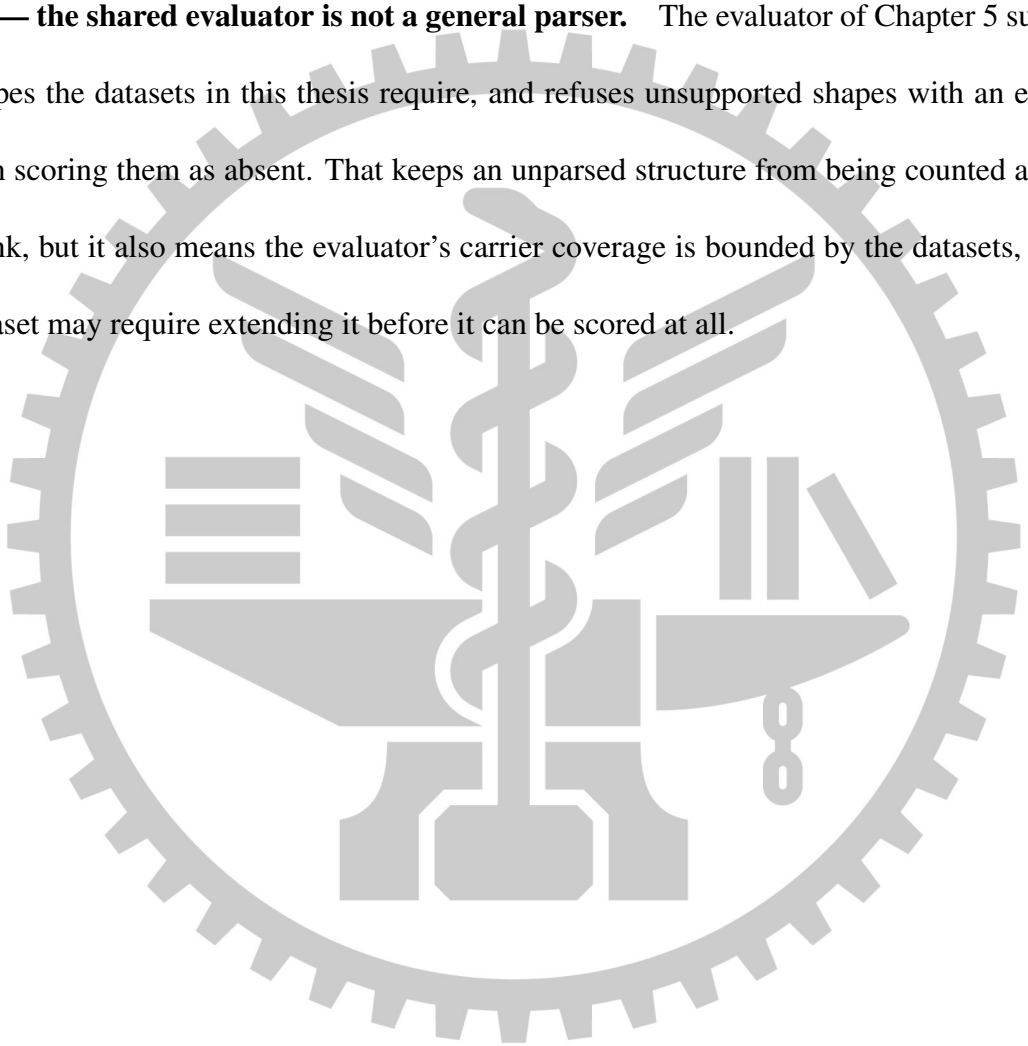
The first is that the unresolved set was computed as "every field without a verified target", which does not distinguish an output field whose target could not be located from an *intermediate* field that legitimately has no target — one carrying a located source value into another field's formula and never written anywhere. The most recent XLSX derivation contained five such fields, each with a working source locator, each read by a header-summary formula, and each recorded as blocking publication. The correction has two halves and the order between them is not optional: intermediate inputs were first admitted to the executable locator set, and only then were they exempted from the unresolved report. The exemption is safe solely because a new hard error now refuses any specification whose formulas read a name that nothing executable can supply. That error was not decoration: replayed across the thirty-five stored specifications it named eleven, and each of the eleven genuinely fails to compile, with no false positive. Before it existed, the two specifications this thesis had recorded as its best both stored a successful validation result and both failed at execution with an unknown formula field.

The second is that two counters reported the same quantity and were never compared. Every narrative of "n unresolved items" in this project's records is the length of the agent's own self-reported list; the publish gate counts a different, canonical set. On the best XLSX specification the two read 13 and 47, and 34 of the 47 were detail cells that the repeat block writes correctly — admitted to the gate's input because the author-time oracle reads expected text from that same list, for a different question. The gate now excludes cells a block owns, recomputed from the specification rather than read from an evidence tag, so that a later version which removes the block puts the same cells back to blocking. No test failed while this was wrong, because what was wrong was a count.

What remains true, and is why D5 stays in this list: after both corrections the XLSX specification is still unpublishable, and decomposing the residue shows the gate is right to block

it. One of the remaining rules requires a value in a column that the specification declares no field for at all, and execution confirms the consequence — three cells scored as silent failures on instances the derivation never saw, because the calibration instance happened to leave that column empty. A count of unresolved items conflates a gate that is too strict with a gate that is doing its job, and the two are repaired in opposite directions.

D6 — the shared evaluator is not a general parser. The evaluator of Chapter 5 supports the shapes the datasets in this thesis require, and refuses unsupported shapes with an error rather than scoring them as absent. That keeps an unparsed structure from being counted as a correct blank, but it also means the evaluator's carrier coverage is bounded by the datasets, and a new dataset may require extending it before it can be scored at all.



Chapter 5. Performance Evaluation

5.1 Research Questions

The claim under test is not that a language model can fill in a form. It is that, for a bounded class of recurring template-instantiation tasks, the model can be removed from the recurring execution path altogether without losing correctness, traceability, or fail-loud behaviour. Four research questions decompose that claim; a fifth asks which components of the derivation procedure carry it.

RQ1 — Correctness. Is a compiled specification's field-level semantic accuracy at least as good as a per-instance agent's on the same instances, measured by the same deterministic evaluator?

RQ2 — Consistency. How much does repeated execution of the same task vary? The question is asymmetric by construction, and the asymmetry is the point: for a per-instance agent the variance is present in every one of the N executions, whereas for the compiled method execution is deterministic and *all* of the variance sits in the single derivation. RQ2 therefore has two halves, and reporting only the first would be a misstatement: run-to-run variation of the baselines, and derivation-to-derivation variation of the compiled specifications.

RQ3 — Cost structure. What is the shape of cost against instance count? A one-off derivation cost plus a near-zero marginal execution cost crosses a per-instance cost at some N; the

question is where that crossing lies, and how sensitive it is to derivation retries.

RQ4 — Behaviour under template drift. When the template changes underneath a frozen artifact, does the method fail loudly and locate the change, and do the baselines fail silently? This is the direct test of the design principle that governs the whole system.

RQ5 — Component contribution. Which of the structural gates, the two oracles, and the repair loop are load-bearing, and which are redundant with each other?

An earlier version of this plan carried a sixth question — whether the method generalizes beyond drawings to office documents. It has been removed rather than answered, because DOCX and XLSX are now the primary evaluation carriers rather than a proof of concept, and a question whose answer is a premise of the study design is not a research question.

A revision to the emphasis, and the measurement that forced it. The original framing put RQ1 first and treated RQ2 and RQ3 as secondary. A pilot measurement on the DOCX dataset changed that ordering. Running a per-instance agent three times over one calibration instance produced semantically correct output every time — the cross-file lookup, the rounding, the gram conversion, and the extent decision were all right — and three different files: three distinct output hashes in three runs, and cell-level agreement of 119 of 172 cells, or 69.2%. All 53 divergent cells were formatting decisions retaken from scratch on each run, a thousands separator present in one run and absent in the next, and none of them was a semantic error.

The consequence is stated here rather than in the results, because it is a claim about what this thesis argues: the primary contribution is not that the compiled method is *more accurate* than a competent agent on tasks a competent agent can do. It is that accuracy is comparable while consistency and cost structure differ structurally. Two tempting responses to that pilot were considered and rejected. Raising task difficulty until the baseline fails would tune the

benchmark towards making the comparator fail; substituting a weaker model would be choosing the opponent. Both are validity threats, and both are recorded as rejected in Section 5.10.

Status: implementation-backed as a measurement, normative as a research plan. The pilot above was executed and its artifacts are retained. The five questions are a plan; the formal experiment has not been run, and no figure in this chapter should be read as an answer to any of them.

5.2 Study Design

5.2.1 Why Not a Few-Shot Held-Out Split

The conventional evaluation for a language-model system is a few-shot held-out split: the model sees K demonstrations and is scored on unseen tasks. That protocol borrows its meaning from in-context learning, and this method is not that shape.

The template is an *input*, not the object of generalization. The system is given the template on purpose; so is every baseline, on every one of its executions. Seeing the template is therefore not leakage but a premise, and it is symmetric across arms. This makes the standard split either vacuous or unfair depending on how it is drawn. If the held-out instances share a template with the demonstrations, nothing was held out that matters: the compiled specification and the agent's demonstrations both describe that template. If the held-out instances come from a different template, the compiled method must derive again — the demonstrations do it no good at all — and the comparison measures the cost of a derivation rather than generalization from examples.

What the split is genuinely needed for is narrower, and is retained: preventing the situation in which one arm's prompts, gates, or intermediate representation were tuned on the very data it

is scored on while the baselines were not. The instrument for that is separation at the *template* level, and Section 5.2.2 reports how far that separation actually holds in this study.

Status: normative, and deliberately so. This subsection argues a design choice. It is included because the objection it answers is the first one a reader of a language-model-systems paper will raise, and because declining a standard protocol without stating why reads as avoiding it.

5.2.2 Two-Level Split

The split has two levels, and only the first of them currently holds in full.

Instance level. Every evaluation template carries ten instances: $K = 3$ calibration instances, shared by all three arms and the only instances any arm may see during construction; two development instances used for debugging; and $M = 5$ sealed evaluation instances that no arm sees at any stage. Two of the five evaluation instances in each dataset are cases whose *correct* behaviour is a refusal, so that an arm which always produces a file cannot score well merely by producing files.

Template level. The evaluation templates are two, with a third planned:

- **S4** (DOCX): a Commercial Invoice and a Packing List produced together from a product master file and a shipment detail file. Its principal difficulty is cross-file computation — unit price from one workbook, quantity from another — and it is the only DOCX evaluation dataset, so every DOCX claim in this thesis rests on it. Thirty-four applicable rules.

- **X1** (XLSX): the Directorate General of Highways monthly site-audit report chain, in

which N real 13020A audit sheets are aggregated into one 13020D monthly report. Its principal difficulties are a cross-file collection of source artifacts and a layout in which one logical record occupies two physical rows. Twenty-seven applicable rules.

- **S5** (XLSX, not yet built): a credit-programme completion audit, chosen for its density of negative obligations.

Two further items must be named, because their absence changes how the split reads. **S3** is a synthetic pilot dataset that was built to protocol and then rejected on the grounds that its layout and content were not those of a document that really circulates; it is retained only as a pilot for the evaluator and the range locator, is not in the main results table, and is not offered as evidence of external validity. **S1** and **S2** appear in the protocol document as a proposed assignment of two development templates and *were never built*.

The gap this leaves, and the direction of its bias. Because S1 and S2 do not exist, development did not happen on templates disjoint from the evaluation set. It happened on the DWG case study, on the rejected pilot, and — decisively — on the calibration instances of S4 and X1 themselves. Three intermediate-representation primitives were added specifically so that the compiled arm could produce X1 at all: artifact collections, stacked records with a slot origin, and equality merge between resolved row sets. X1 is an evaluation template. The template-level separation the protocol describes is therefore, at present, a normative claim rather than a property of this study, and its bias runs *in favour of the proposed method*.

Following the third of the three admissible responses to such a gap, the claim is kept and labelled rather than quietly weakened, and the bias is disclosed again in Section 5.10. Two measurements bound its size without cancelling it. First, each primitive was added only after it had been observed to fail on a real file, and its collateral effect on the already-stored specifications

was swept: the slot-alignment gate introduced with the third primitive was replayed against all thirty-four stored specifications and matched exactly the two it was written for, with no false positive on the five specifications of the other dataset. Second, value-level verification of the resulting specification was performed on development instances the derivation never saw: the fourth X1 specification reproduced 255 of 255 scored values across the three calibration and two development instances using the scoring policy's own comparator, and a deliberately corrupted variant of the same specification scored 240. Neither measurement makes the development location correct; they bound how far it could have been exploited.

A second construct-validity item, recorded here and again in Section 5.10. X1's rules require projects to be reported in ascending project-number order. The real 13020D form prescribes no ordering; the ordering is a convention this dataset introduced so that the extent has a checkable manifestation. The mechanism that implements it is built and tested, but the question of *who decided the order* belongs to dataset construct validity rather than to the method.

Status: implementation-backed at the instance level, normative at the template level, with the bias direction stated.

5.2.3 Dataset Construction

5.3 Baselines and Fairness

Three arms are compared. They differ only in what persists between instances.

Arm A — per-instance agent. For each instance, an agent is given the template, the rule document, and the source data, and produces the output document. Nothing persists: the next instance starts from the same prompt with no artifact carried over. This is the status quo

the thesis argues against, and it is deliberately a strong version of it — a current frontier model with the same structured inspection tools the proposed method uses.

Arm B — agent-built frozen skill. The agent is first allowed to generalize from the calibration instances into a reusable artifact of its own design — a script, a checklist, a set of instructions — which is then frozen. Every evaluation instance is executed by an agent that may read that artifact and may not modify it.

Arm C — the proposed method. One derivation produces a specification; every instance is executed by the deterministic compiler with no model call at all.

Why three arms and not two. Arm B exists to answer the strongest available objection to the result. If only A and C were compared, any advantage of C could be attributed to the mere existence of a persistent artifact rather than to anything about how that artifact is produced and verified. Arm B holds persistence fixed and varies only the production procedure: an agent generalizing freely, versus a derivation constrained by an intermediate representation, structural gates, and two oracles. The question Arm B answers is therefore the one that actually distinguishes the contribution — is the verified derivation better than an agent's own attempt at the same idea?

Fairness has two layers, and the second is the one usually omitted. The first layer is *tooling parity*: the same structured dump utilities, the same model, the same retry ceiling, the same wall-clock ceiling. The second is *material parity*: Arms A and B receive an input list that is byte-identical to the one Arm C's derivation receives, because an advantage produced by handing one arm a cleaner or richer set of inputs is not an advantage of the method. The frozen input manifest is truth-free by construction and its envelope is hashable, so what each arm was given is recorded rather than asserted.

Material parity has a specific consequence that must be stated because it works against the proposed method. The rule material given to any arm may not contain target coordinates — no cell addresses, no bookmarks, no content-control tags, no drawing handles. Arm C's derivation therefore has to locate its own write targets, and cannot be given them; the rule table is not permitted to become an answer key for anyone.

Arm B's freeze must be enforced by the filesystem. An instruction not to modify the skill is not a control. The frozen artifact is hashed and mounted read-only for the evaluation phase; an attempt to write to it is recorded as a protocol violation and reported rather than silently tolerated. This matters because the failure it prevents — an agent quietly improving its own skill between instances — would convert Arm B into a per-instance arm while still being reported as a frozen one.

Status: contract implementation-backed, live execution not yet built. The three-arm input and evidence contract is frozen and tested: the truth-free input manifest, the hashable envelope, the Arm B freeze, namespace isolation for Arm C, retention of raw attempt evidence, and fake-capture negative controls are all implemented. The live adapters that actually drive Arms A and B, and the formal wrapper around Arm C, are not. The pilot reported in Section 5.1 was run outside that harness and is labelled as a pilot for exactly that reason; it must not be read as an Arm A result.

5.4 Metrics

5.4.1 Asymmetric Rerun Budget

5.5 The Evaluator

All three arms are scored by one deterministic program. It parses the produced artifact, extracts the values at the locations the dataset's annotation manifest names, compares them against that instance's ground truth, and emits per-check results. It never asks a language model whether an output looks right, for the reason given in Section 4.3: a judge that can be wrong in the same direction as the system under test cannot bound the system's error.

5.5.1 Separate Channels, Never One Average

Scores are partitioned into channels, each with its own denominator: *semantic* (the primary metric), *extent* (which source rows were included), *refusal* (a case whose correct behaviour is to refuse), *structure* (the artifact opens and has the expected shape), *presentation* (formatting, reported separately from meaning), *untouched* (regions no field declares were not modified), and *integrity* (the package's critical parts survived). A check whose prefix is not one of the declared channels fails closed rather than being scored as absent. The policy is versioned, and every result record carries the evaluator version, the policy version, the source hash of the scoring code, and the repository commit.

The partition is not presentational. The one measurement that settles it: while the "this slot must remain blank" obligations were counted inside the semantic denominator, an arm that submitted the blank template unchanged — writing nothing whatsoever — scored 66.7% semantic correctness on one instance. After the obligations were moved into their own channel

the same output scores 0%. Every individual check had been correct; what was wrong was which total they were added into.

5.5.2 Two Self-Validations, Neither Sufficient Alone

Forward. The evaluator must score the generator's own expected artifacts at 100%. This catches an evaluator that reads the wrong location or compares with the wrong tolerance.

Reverse. Controlled corruptions are applied to the expected artifact and the evaluator must catch every one. Without this, an evaluator that always answers "pass" would give every arm a perfect score and the forward test would not notice. The current mutation coverage is 6 of 6 on the pilot dataset, 12 of 12 on S4, and 13 of 13 on X1; X1 is the first dataset in which *all seven* channels have at least one mutation that exercises them, which closed two channels that had previously been structurally vacuous.

A mutation test is not a validity proof, and the gap is specific. The expected artifact and the ground truth are produced by one generator. If the generator computes a value wrongly, both are wrong together and the evaluator faithfully reports 100%. Mutation testing establishes that the evaluator detects corruption; it says nothing about whether the generator computed correctly. The two do not overlap. The present detection mechanism for that residual is a per-dataset spot audit — at least three fields per dataset recomputed by hand from the rule document, including one aggregate and one negative obligation — which is probabilistic rather than a guarantee, and is reported as such.

5.5.3 A Negative Control for Every Zero

The evaluator and the probes around it are themselves software, and the failure mode observed repeatedly during construction is that a broken probe reports a serene zero: zero mismatches, zero problems, full marks. Measured instances during this work include a probe that used the wrong key name, one that omitted a worksheet prefix from every reference so that nothing matched anything, one whose trial compilation failed so that the comparison returned an empty list, and one that compared string forms where the annotation declared exact decimal comparison, so that twenty *correct* cells were reported as wrong. Two of those report "looks broken" and the rest report "looks fine"; the second kind is the dangerous one, and it is indistinguishable from success by inspection.

The rule adopted is therefore mechanical rather than advisory. Any reported zero or perfect score must be accompanied by a deliberate control that makes the figure move, and the control must distinguish doing the work from not doing it — a control whose expected value also happens to be the result of skipping the mechanism measures nothing and never turns red. That last clause is not hypothetical: an ordering test was found whose expected row order equalled the arrival order, so disabling the sort entirely left it green.

5.5.4 The Extent Diagnostic Is Not Symmetric Across Arms

Only Arm C declares which source rows it included; Arms A and B produce a document and no provenance. Scoring the extent channel from an arm's own declaration would repeat exactly the mistake that the coverage denominator made in Section 4.6 — a figure self-reported by the system being scored is not a cross-arm metric.

The cross-arm instrument is therefore a set of *cross-checks*: identities between two independently scored fields of the same instance, which any arm's output either satisfies or does

not. Alongside them, each dataset's manifest declares, for every wrong-extent hypothesis, the values that hypothesis would produce, so a wrong extent can be *named* from the output alone rather than inferred. The effect is measurable: an output that occupies one extra detail slot while leaving all six totals byte-identical passes all 112 semantic checks on one instance and is still identified as a specific wrong-extent hypothesis. Arm C's own resolved anchors are reported as supplementary evidence and are explicitly marked as available for one arm only; they are never placed in a column alongside A and B.

Status: implementation-backed. The shared evaluator lints, self-tests, and scores all three datasets; the scoring policy is frozen at version 1.1.0; the mutation suites and their per-channel negative controls are in the repository and run in continuous integration. What is not established is the generator's own arithmetic, as stated above.

5.6 Template Drift

5.7 Ablations

5.8 Results

5.8.1 Where the Randomness Went

5.9 Case Study: Engineering Drawings

5.10 Threats to Validity

Each item below states what is threatened, what was measured, and which arm the bias favours. Items whose direction favours the proposed method are listed first, because those are the ones a reader has no independent way to discover.

5.10.1 Threats Biased Towards the Proposed Method

T1 — development happened on evaluation templates. The template-level separation described in Section 5.2.2 does not hold: the two development templates the protocol proposes were never built, and the prompts, gates, and intermediate representation were tuned against the calibration instances of the evaluation templates themselves. Three representation primitives exist specifically because the compiled arm could not otherwise produce X1. The bias favours Arm C, and it is not quantified — the only way to quantify it is to build the missing development templates. The two bounding measurements are given in Section 5.2.2.

T2 — the cell-datatype exemption. The compiler stringifies every value before writing it, so a numeric field lands in a spreadsheet as text. The annotation rules therefore declare `presentation.dat` as `any` for every field. Without that exemption Arm C would lose a point on every numeric field, and the study would be measuring a write-side type policy rather than the method. With it, a real defect — numbers stored as text — is not scored, and a baseline that writes correct types earns nothing for doing so. This is the third admissible response to a gap: the claim is kept, labelled normative, and the direction disclosed. It favours Arm C.

T3 — the rule tables are ours. Even where the template is a genuine third-party artifact, the machine-readable rule applicability manifest that supplies the coverage denominator was written for this study. It weakens the objection that the templates are self-designed; it does not weaken the objection that the rules were written by someone who knew what the method can express. Any use of an externally sourced template must state this in the same breath, or it reads as padding.

T4 — three primitives rest on a single example. Artifact collections, stacked records, and equality merge between row sets were each validated on one dataset, and it is the same dataset in all three cases. The key–value extent added for the DOCX dataset has the same property: a survey of six real government forms found no second instance of the shape. The failure mode this predicts is not a red test but a different form that silently does not fit, and no measurement in this thesis would detect it.

5.10.2 Threats Biased Against the Proposed Method

T5 — package preservation was, until recently, unmeasurable for Arm C. The integrity channel treats drawings and printer settings as critical parts of a spreadsheet package, which

is why the government form was chosen as a dataset in the first place. Until it was corrected, the production write-back for XLSX serialized the workbook through a spreadsheet model, and that operation drops exactly those two parts — the same two parts the mutation suite's integrity negative control drops, because the negative control *is* a round trip through the same library. Arm C therefore scored 0 of 5 on integrity by construction, and the channel could not distinguish a correct run from a deliberately corrupted one.

The response was the first admissible one: the write-back was replaced by a package editor that edits the one worksheet part that must change and copies every other byte through. The measurement after that change, on the same five instances, is 5 of 5 with no preservation violation, while re-serializing the same outputs through the spreadsheet model still yields 0 of 5. Narrowing the channel to fit the writer was considered and rejected, because it would have been a reduction of the thesis claim to match the implementation. The threat is recorded rather than deleted, because every integrity figure produced before that correction is an artifact of the writer and not a result.

T6 — the DOCX write path has not had the same audit. The correction in T5 covers XLSX only. The layout-bearing parts of a Word package have not been enumerated part by part, and S4 is the only DOCX evaluation dataset. Any DOCX integrity figure in this thesis therefore carries an unmeasured residual.

T7 — an unpublishable specification is not necessarily a failed one. The publish gate blocks on any unresolved item, and the unresolved set has repeatedly been found to contain entries that are not defects. Two counters existed for the same quantity — the agent's own self-reported list and the gate's canonical set — and they were never compared: on the best X1 specification they read 13 and 47. Thirty-four of the 47 were detail cells that a repeat block

writes correctly, admitted to the gate's input list because the author-time oracle reads expected text from that same list for a different purpose. Any count of unresolved items must be decomposed before it is interpreted, and a headline "unpublishable" figure without that decomposition would understate the method.

5.10.3 Threats of Undetermined or Mixed Direction

T8 — dataset scale. Two evaluation templates exist and a third is planned, against an original intention of three to five per carrier. Conclusions must be descriptive; no statistical significance may be claimed, and no single average may be taken across carriers.

T9 — the inclusion criterion is itself skewed. The datasets admit only tasks whose target row count is fixed, because the write side deliberately does not insert rows, and they exclude formula results. Both exclusions coincide with boundaries of the proposed method, so the excluded region may contain tasks the baselines can do and the compiled method cannot. This is the second admissible response — an explicit expressiveness boundary — and it is disclosed rather than argued away.

T10 — what the untouched channel does not reach. Untouched-region comparison covers cells and paragraphs. It does not reach footnotes, endnotes, comments, or other independent package parts, and styles and formula results are outside the first version of the scoring policy by design, to avoid confounding correctness with whether a spreadsheet application refreshed its cached values.

T11 — model dependence. Every derivation reported here used one model family. The ablation in Section 5.7 is the instrument for this and has not been run; until it is, no claim in this thesis is established as vendor-independent.

T12 — the correctness of the reference outputs. Discussed in Section 5.5.2. Mutation testing bounds the evaluator, not the generator; the residual is covered only by a probabilistic spot audit.

T13 — two responses to the pilot that were rejected, and why. After the pilot in Section 5.1 showed the per-instance baseline to be semantically accurate, two adjustments were available and both were declined. Increasing task difficulty until the baseline begins to fail would tune the benchmark towards producing the desired comparison, and substituting a weaker model would amount to selecting the opponent. Recording them here is part of the claim: the emphasis of this thesis moved to consistency and cost structure *because* the baseline was accurate, not because the tasks were made harder.

Status: implementation-backed for T1, T2, T5, T6, T7, T9, T10, T12; normative for T3, T4, T8, T11, T13. The items marked implementation-backed each name a measurement in this repository. The others are stated positions or unrun experiments, and are labelled so that no reader takes an intention for a result.

Chapter 6. Conclusions

6.1 Conclusion

6.2 Limitations and When This Method Does Not Apply

6.3 Future Work

References